Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** To install and configure Flutter Environment. Create an app to display "Hello Username" and on press button to display a greeting message.
Experiment No:- 01	**Date:-** 19-09-2024 / **Enrollment No:-** 92200133030

Objective:- To install and configure Flutter Environment. Create an app to display "Hello Username" and on press button to display a greeting message.

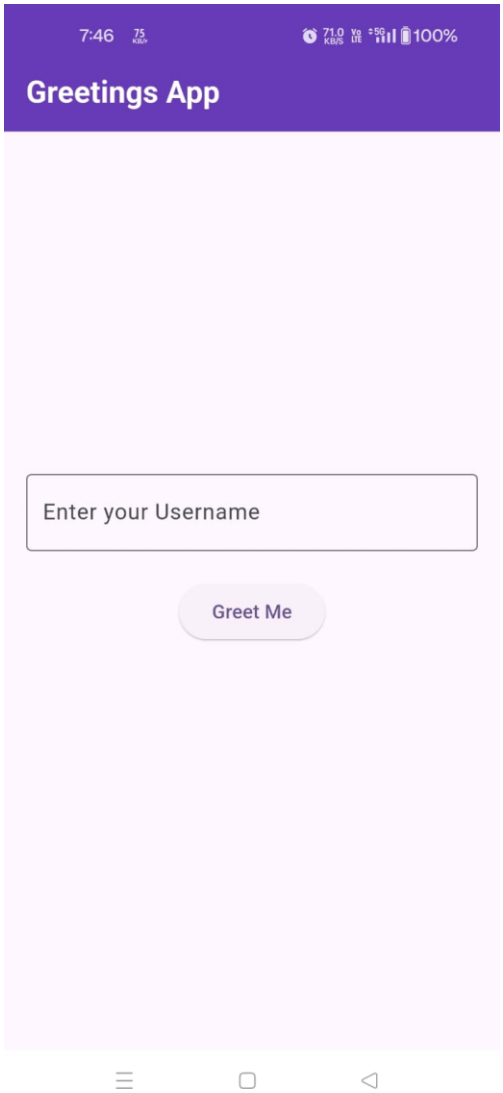
Code :-

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Demo',
      debugShowCheckedModeBanner: false, // Remove the debug ribbon from the app
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
        useMaterial3: true,
      ),
      home: GreetingScreen(), // The home screen of the app is GreetingScreen
    );
  }
}

// GreetingScreen - A stateful widget as it will manage dynamic state
class GreetingScreen extends StatefulWidget {
  @override
  _GreetingScreenState createState() => _GreetingScreenState();
}

// State class for GreetingScreen
class _GreetingScreenState extends State<GreetingScreen> {
  String username = ''; // Variable to store the username entered by the user
  String greetingMessage =
      ''; // Variable to store the greeting message generated

  TextEditingController controller =
      TextEditingController(); // Controller for managing the TextField input

  // Function to generate a greeting message based on the input username
  void generateGreeting() {
    setState(() {
      greetingMessage =
          'Hello $username, Greetings of the day!'; // Update greeting message
    });
  }
```

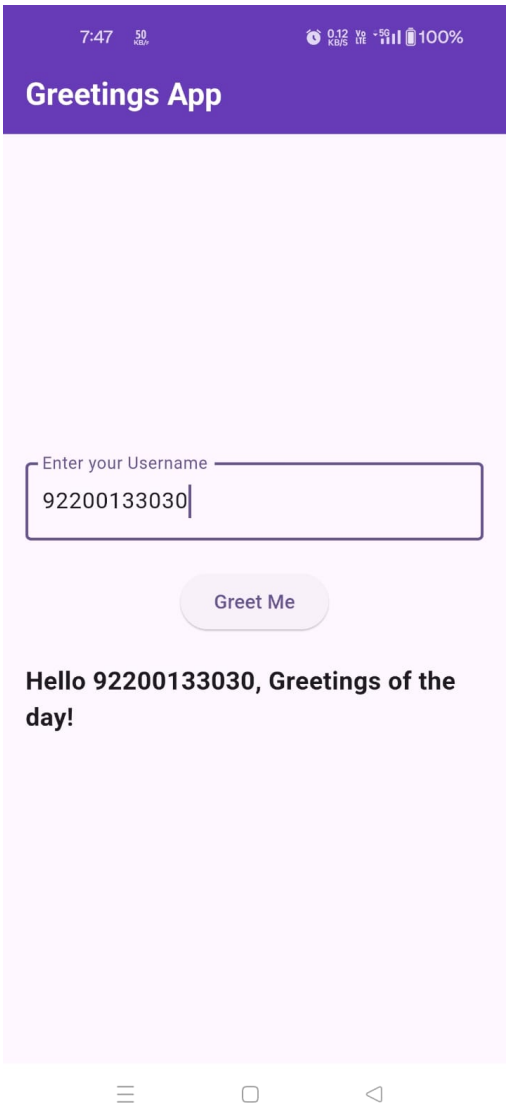
```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text(
        'Greetings App',
        style: TextStyle(
          color: Colors.white,
          fontWeight: FontWeight.bold,
        ),
      ),
      backgroundColor: Colors.deepPurple, // Set the app bar background color
    ),
    body: Padding(
      padding: const EdgeInsets.all(16.0), // Add padding to the content
      child: Column(
        mainAxisAlignment:
            MainAxisAlignment.center, // Center the content vertically
        children: <Widget>[
          // TextField for entering the username
          TextField(
            controller: controller, // Connect controller to capture the input
            decoration: const InputDecoration(
              labelText: 'Enter your Username', // Label for the text field
              border:
                  OutlineInputBorder(), // Add an outline border around the text field
            ),
            onChanged: (text) {
              // Update the username as the text field changes
              username = text;
            },
          ),
          const SizedBox(height: 20), // Add some space between the widgets
          // Button to trigger the greeting generation
          ElevatedButton(
            onPressed:
                generateGreeting, // Call the generateGreeting function when pressed
            child: const Text('Greet Me'),
          ),
          const SizedBox(height: 20), // Add space between button and message
          // Display the generated greeting message
          Text(
            greetingMessage,
            style: const TextStyle(
                fontSize: 18,
                fontWeight:
                    FontWeight.bold), // Style for the greeting message
          ),
        ],
      ),
    ),
```

```
    );
  }
}
```

Output:-


7:46
100%
Greetings App
Enter your Username
Greet Me

7:47
100%
Greetings App
Enter your Username
92200133030
Greet Me


7:47
100%
Greetings App
Enter your Username
92200133030
Greet Me
Hello 92200133030, Greetings of the day!

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** To Study the basics of Dart language and design a basic Flutter App to Create Unit convertor(for length, weight, temperature, Area) app using Flutter.
Experiment No:- 02	**Date:-** 20-09-2024 / **Enrollment No:-** 92200133030

Objective:- To Study the basics of Dart language and design a basic Flutter App to Create Unit convertor(for length, weight, temperature, Area) app using Flutter.

Code :-

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Demo',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
        useMaterial3: true,
      ),
      home: ConverterScreen(),
    );
  }
}

class ConverterScreen extends StatefulWidget {
  @override
  _ConverterScreenState createState() => _ConverterScreenState();
}

class _ConverterScreenState extends State<ConverterScreen> {
  // Conversion options with units (from -> to)
  final List<String> _conversionTypes = [
    'Meters to Kilometers',
    'Kilometers to Meters',
    'Grams to Kilograms',
    'Kilograms to Grams',
    'Celsius to Fahrenheit',
    'Fahrenheit to Celsius',
    'Square Meters to Hectares',
    'Hectares to Square Meters',
    // New conversions
    'Miles to Kilometers',
    'Kilometers to Miles',
    'Pounds to Kilograms',
    'Kilograms to Pounds',
    'Liters to Milliliters',
```

```
  'Milliliters to Liters',
  'Kilometers per hour to Miles per hour',
  'Miles per hour to Kilometers per hour'
];

// Mapping for input and output units
final Map<String, String> _inputUnits = {
  'Meters to Kilometers': 'meters',
  'Kilometers to Meters': 'kilometers',
  'Grams to Kilograms': 'grams',
  'Kilograms to Grams': 'kilograms',
  'Celsius to Fahrenheit': 'Celsius',
  'Fahrenheit to Celsius': 'Fahrenheit',
  'Square Meters to Hectares': 'square meters',
  'Hectares to Square Meters': 'hectares',
  'Miles to Kilometers': 'miles',
  'Kilometers to Miles': 'kilometers',
  'Pounds to Kilograms': 'pounds',
  'Kilograms to Pounds': 'kilograms',
  'Liters to Milliliters': 'liters',
  'Milliliters to Liters': 'milliliters',
  'Kilometers per hour to Miles per hour': 'km/h',
  'Miles per hour to Kilometers per hour': 'mph',
};

final Map<String, String> _outputUnits = {
  'Meters to Kilometers': 'kilometers',
  'Kilometers to Meters': 'meters',
  'Grams to Kilograms': 'kilograms',
  'Kilograms to Grams': 'grams',
  'Celsius to Fahrenheit': 'Fahrenheit',
  'Fahrenheit to Celsius': 'Celsius',
  'Square Meters to Hectares': 'hectares',
  'Hectares to Square Meters': 'square meters',
  'Miles to Kilometers': 'kilometers',
  'Kilometers to Miles': 'miles',
  'Pounds to Kilograms': 'kilograms',
  'Kilograms to Pounds': 'pounds',
  'Liters to Milliliters': 'milliliters',
  'Milliliters to Liters': 'liters',
  'Kilometers per hour to Miles per hour': 'mph',
  'Miles per hour to Kilometers per hour': 'km/h',
};

// Input controller
final TextEditingController _inputController = TextEditingController();

// Selected conversion type and result
String _selectedConversion = 'Meters to Kilometers';
double _result = 0.0;

@override
```

```
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: Text(
        'Unit Converter',
        style: TextStyle(
          color: Colors.white,
          fontWeight: FontWeight.bold,
        ),
      ),
      backgroundColor: Colors.deepPurple,
    ),
    body: Padding(
      padding: const EdgeInsets.all(16.0),
      child: Column(
        children: <Widget>[
          // Dropdown for selecting conversion type
          DropdownButton<String>(
            value: _selectedConversion,
            onChanged: (String? newValue) {
              setState(() {
                _selectedConversion = newValue!;
              });
            },
            items: _conversionTypes.map((String value) {
              return DropdownMenuItem<String>(
                value: value,
                child: Text(value),
              );
            }).toList(),
          ),
          SizedBox(height: 20),

          // Input field with dynamic unit label
          Row(
            children: [
              Expanded(
                child: TextField(
                  controller: _inputController,
                  keyboardType: TextInputType.number,
                  decoration: InputDecoration(
                    labelText: 'Enter value',
                  ),
                ),
              ),
              SizedBox(width: 10),
              Text(
                _inputUnits[
                    _selectedConversion]!, // Dynamic unit next to input field
                style: TextStyle(fontSize: 16),
              ),
            ],
```

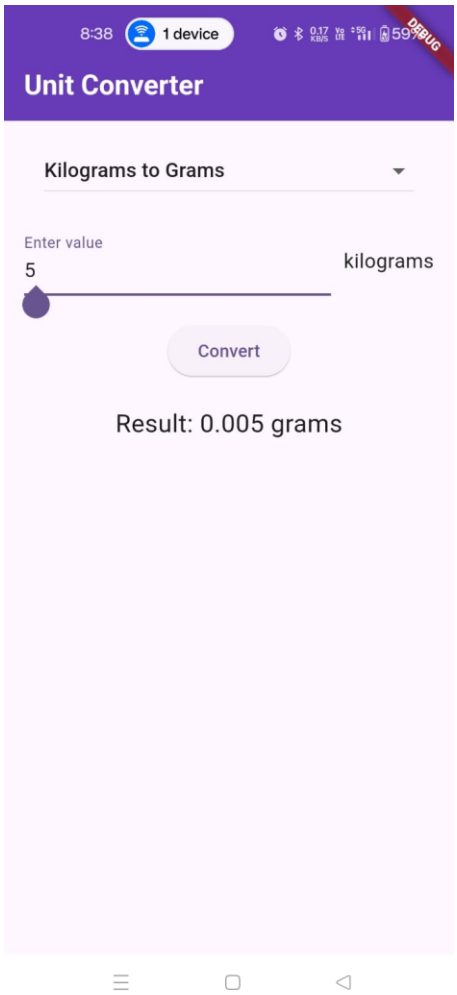
```
          ),
          SizedBox(height: 20),

          // Convert button
          ElevatedButton(
            onPressed: _convert,
            child: Text('Convert'),
          ),
          SizedBox(height: 20),

          // Display result with unit
          Text(
            'Result: $_result ${_outputUnits[_selectedConversion]}',
            style: TextStyle(fontSize: 20),
          ),
        ],
      ),
    ),
  );
}

// Conversion logic based on selected type
void _convert() {
  double inputValue = double.tryParse(_inputController.text) ?? 0.0;

  switch (_selectedConversion) {
    case 'Meters to Kilometers':
      _result = _convertMetersToKilometers(inputValue);
      break;
    case 'Kilometers to Meters':
      _result = _convertKilometersToMeters(inputValue);
      break;
    case 'Grams to Kilograms':
      _result = _convertGramsToKilograms(inputValue);
      break;
    case 'Kilograms to Grams':
      _result = _convertKilogramsToGrams(inputValue);
      break;
    case 'Celsius to Fahrenheit':
      _result = _convertCelsiusToFahrenheit(inputValue);
      break;
    case 'Fahrenheit to Celsius':
      _result = _convertFahrenheitToCelsius(inputValue);
      break;
    case 'Square Meters to Hectares':
      _result = _convertSquareMetersToHectares(inputValue);
      break;
    case 'Hectares to Square Meters':
      _result = _convertHectaresToSquareMeters(inputValue);
      break;
    // New conversions
    case 'Miles to Kilometers':
```

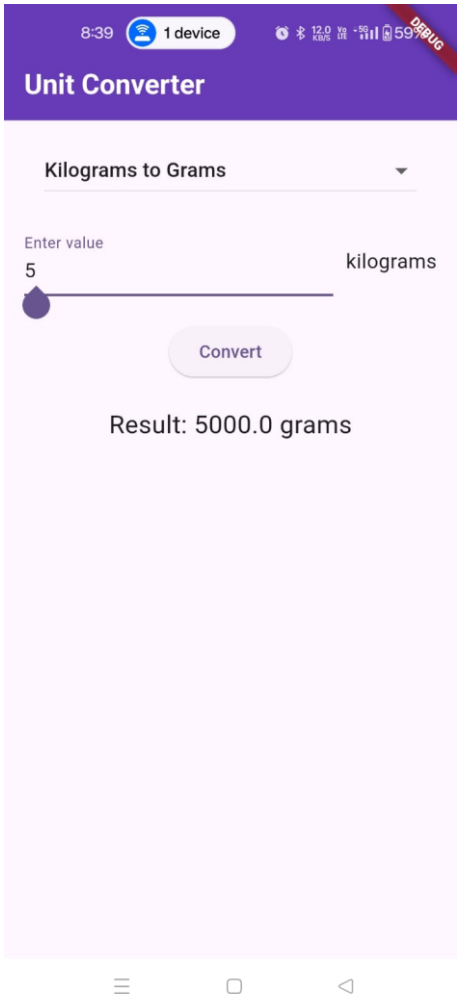
```
      _result = _convertMilesToKilometers(inputValue);
      break;
    case 'Kilometers to Miles':
      _result = _convertKilometersToMiles(inputValue);
      break;
    case 'Pounds to Kilograms':
      _result = _convertPoundsToKilograms(inputValue);
      break;
    case 'Kilograms to Pounds':
      _result = _convertKilogramsToPounds(inputValue);
      break;
    case 'Liters to Milliliters':
      _result = _convertLitersToMilliliters(inputValue);
      break;
    case 'Milliliters to Liters':
      _result = _convertMillilitersToLiters(inputValue);
      break;
    case 'Kilometers per hour to Miles per hour':
      _result = _convertKilometersToMilesPerHour(inputValue);
      break;
    case 'Miles per hour to Kilometers per hour':
      _result = _convertMilesToKilometersPerHour(inputValue);
      break;
    default:
      _result = 0.0;
  }

  setState(() {});
}

// Conversion functions for each type
// Length conversions
double _convertMetersToKilometers(double value) {
  return value / 1000; // 1 meter = 0.001 kilometers
}

double _convertKilometersToMeters(double value) {
  return value * 1000; // 1 kilometer = 1000 meters
}

// Weight conversions
double _convertGramsToKilograms(double value) {
  return value / 1000; // 1 gram = 0.001 kilograms
}

double _convertKilogramsToGrams(double value) {
  return value * 1000; // 1 kilogram = 1000 grams
}

// Temperature conversions
double _convertCelsiusToFahrenheit(double value) {
  return (value * 9 / 5) + 32; // (C * 9/5) + 32 = F
```

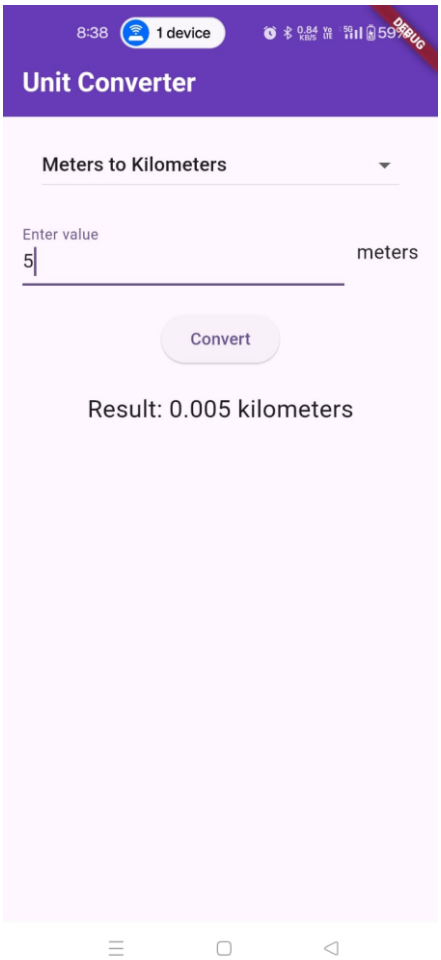
```
  }

  double _convertFahrenheitToCelsius(double value) {
    return (value - 32) * 5 / 9; // (F - 32) * 5/9 = C
  }

  // Area conversions
  double _convertSquareMetersToHectares(double value) {
    return value / 10000; // 1 square meter = 0.0001 hectares
  }

  double _convertHectaresToSquareMeters(double value) {
    return value * 10000; // 1 hectare = 10000 square meters
  }

  // New conversion functions
  // Length: Miles and Kilometers
  double _convertMilesToKilometers(double value) {
    return value * 1.60934; // 1 mile = 1.60934 kilometers
  }

  double _convertKilometersToMiles(double value) {
    return value / 1.60934; // 1 kilometer = 0.621371 miles
  }

  // Weight: Pounds and Kilograms
  double _convertPoundsToKilograms(double value) {
    return value * 0.453592; // 1 pound = 0.453592 kilograms
  }

  double _convertKilogramsToPounds(double value) {
    return value / 0.453592; // 1 kilogram = 2.20462 pounds
  }

  // Volume: Liters and Milliliters
  double _convertLitersToMilliliters(double value) {
    return value * 1000; // 1 liter = 1000 milliliters
  }

  double _convertMillilitersToLiters(double value) {
    return value / 1000; // 1 milliliter = 0.001 liters
  }

  // Speed: Kilometers per hour and Miles per hour
  double _convertKilometersToMilesPerHour(double value) {
    return value * 0.621371; // 1 km/h = 0.621371 mph
  }

  double _convertMilesToKilometersPerHour(double value) {
    return value / 0.621371; // 1 mph = 1.60934 km/h
  }
}
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** To Study the basics of Dart language and design a basic Flutter App to Create Unit convertor(for length, weight, temperature, Area) app using Flutter.	
Experiment No:- 02	**Date:-** 20-09-2024	**Enrollment No:-** 92200133030

Output:-







Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** To design Flutter UI by including common widgets, Design BMI Calculator as per given screenshot	
Experiment No:- 03	**Date:-** 26-09-2024	**Enrollment No:-** 92200133030

Objective:- To design Flutter UI by including common widgets, Design BMI Calculator as per given screenshot.

Code :-

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp()); // Entry point of the app
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      theme: ThemeData(
        brightness: Brightness.light, // Light theme settings
        primaryColor: Colors.blue,
      ),
      darkTheme: ThemeData(
        brightness: Brightness.dark, // Dark theme settings
        scaffoldBackgroundColor:
            const Color(0xFF121212), // Background for Scaffold in dark mode
        appBarTheme: const AppBarTheme(
          backgroundColor: Color(0xFF1F1F1F), // Dark app bar color
          iconTheme:
              IconThemeData(color: Colors.white), // Icon color in app bar
          titleTextStyle: TextStyle(color: Colors.white, fontSize: 20),
        ),
        textButtonTheme: TextButtonThemeData(
          style: TextButton.styleFrom(
            foregroundColor: Colors.white,
            backgroundColor: Colors.blue, // TextButton background color
          ),
        ),
        inputDecorationTheme: InputDecorationTheme(
          filled: true,
          fillColor:
              Colors.grey[800], // Input field background color in dark mode
          border: OutlineInputBorder(
            borderRadius: BorderRadius.circular(8.0),
            borderSide: BorderSide.none, // No border for input fields
          ),
          hintStyle: const TextStyle(color: Colors.grey), // Hint text style
        ),
        textTheme: const TextTheme(
          bodyMedium: TextStyle(color: Colors.white), // Body text color
          titleLarge: TextStyle(color: Colors.white), // Large title text color
```

```
      ),
    ),
    themeMode: ThemeMode.dark, // Sets the dark theme as the default
    home: const BMICalculator(), // Home widget is the BMICalculator
    debugShowCheckedModeBanner: false, // Disable the debug banner
  );
 }
}

class BMICalculator extends StatefulWidget {
  const BMICalculator({super.key});

  @override
  State<BMICalculator> createState() => _BMICalculatorState();
}

class _BMICalculatorState extends State<BMICalculator> {
  int currentindex = 0; // Track selected gender
  String result = ""; // Variable to store BMI result
  double height = 0.0; // User height
  double weight = 0.0; // User weight
  TextEditingController heightController =
      TextEditingController(); // Controller for height input
  TextEditingController weightController =
      TextEditingController(); // Controller for weight input

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("BMI Calculator"),
        backgroundColor: Theme.of(context).appBarTheme.backgroundColor,
        actions: [
          IconButton(
            onPressed: () {}, // Placeholder for settings action
            icon: const Icon(Icons.settings),
          ),
        ],
      ),
      body: SingleChildScrollView(
        child: Padding(
          padding: const EdgeInsets.all(12.0), // Padding for content
          child: Column(
            mainAxisAlignment: MainAxisAlignment.start,
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Row(
                children: [
                  radioButton("Man", Colors.blue, 0), // Male option
                  radioButton("Women", Colors.pink, 1), // Female option
                ],
              ),
```

```
const SizedBox(height: 20.0),
const Text(
  "Your Height in Centimeter :- ",
  style: TextStyle(fontSize: 18.0), // Label for height input
),
const SizedBox(height: 8.0),
TextField(
  controller: heightController,
  keyboardType: TextInputType.number,
  textAlign: TextAlign.center,
  decoration: InputDecoration(
    hintText:
        "Your Height in Centimeter", // Placeholder for height input
    filled: true,
    fillColor: Theme.of(context).inputDecorationTheme.fillColor,
    border: OutlineInputBorder(
      borderRadius: BorderRadius.circular(8.0),
      borderSide: BorderSide.none, // No border for height input
    ),
  ),
),
const SizedBox(height: 20.0),
const Text(
  "Your Weight in Kilogram :- ",
  style: TextStyle(fontSize: 18.0), // Label for weight input
),
const SizedBox(height: 8.0),
TextField(
  controller: weightController,
  keyboardType: TextInputType.number,
  textAlign: TextAlign.center,
  decoration: InputDecoration(
    hintText:
        "Your Weight in Kilogram", // Placeholder for weight input
    filled: true,
    fillColor: Theme.of(context).inputDecorationTheme.fillColor,
    border: OutlineInputBorder(
      borderRadius: BorderRadius.circular(8.0),
      borderSide: BorderSide.none, // No border for weight input
    ),
  ),
),
const SizedBox(height: 20.0),
SizedBox(
  width: double.infinity,
  height: 50.0,
  child: TextButton(
    onPressed: () {
      setState(() {
        // Parsing the input values
        height = double.parse(heightController.value.text);
        weight = double.parse(weightController.value.text);
```

```
                  });
                  calculateBMI(height, weight); // Trigger BMI calculation
                },
                child: const Text(
                  "Calculate",
                  style: TextStyle(color: Colors.white),
                ),
              ),
            ),
            const SizedBox(height: 20.0),
            const Center(
              child: Text(
                "Your BMI is :- ",
                textAlign: TextAlign.center, // Label for BMI result
                style: TextStyle(
                  fontSize: 24.0,
                  fontWeight: FontWeight.bold,
                ),
              ),
            ),
            const SizedBox(height: 50.0),
            Center(
              child: Text(
                result, // Display calculated BMI
                textAlign: TextAlign.center,
                style: const TextStyle(
                  fontSize: 40.0,
                  fontWeight: FontWeight.bold,
                ),
              ),
            ),
          ],
        ),
      ),
    ),
  );
}

// Function to calculate BMI based on height and weight
void calculateBMI(double height, double weight) {
  double finalresult = weight / (height * height / 10000); // Formula for BMI
  String bmi =
      finalresult.toStringAsFixed(2); // Format result to 2 decimal places
  setState(() {
    result = bmi; // Update result in the UI
  });
}

// Function to change the selected gender index
void changeIndex(int index) {
  setState(() {
    currentindex = index;
```

```
    });
  }

  // Reusable radio button widget for gender selection
  Widget radioButton(String value, Color color, int index) {
    return Expanded(
      child: Container(
        margin: const EdgeInsets.symmetric(horizontal: 12.0),
        height: 80.0,
        child: TextButton(
          style: TextButton.styleFrom(
            backgroundColor: currentindex == index ? color : Colors.grey[800],
            shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(8.0),
            ),
          ),
          onPressed: () {
            changeIndex(index); // Update selected gender
          },
          child: value == "Man"
              ? Icon(
                  Icons.male,
                  color: currentindex == index ? Colors.white : color,
                  size: 40.0,
                )
              : Icon(
                  Icons.female,
                  color: currentindex == index ? Colors.white : color,
                  size: 40.0,
                ),
        ),
      ),
    );
  }
}
```

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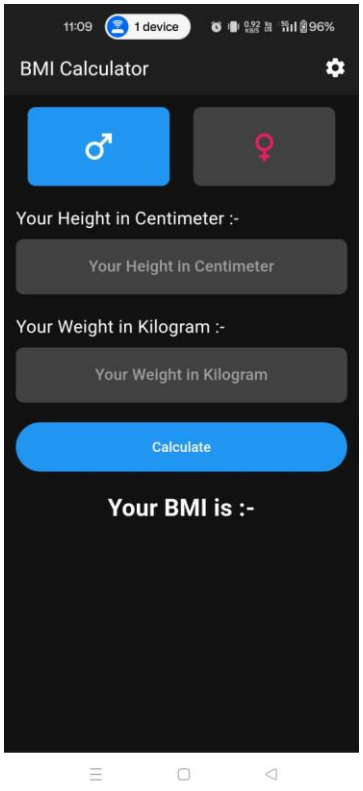
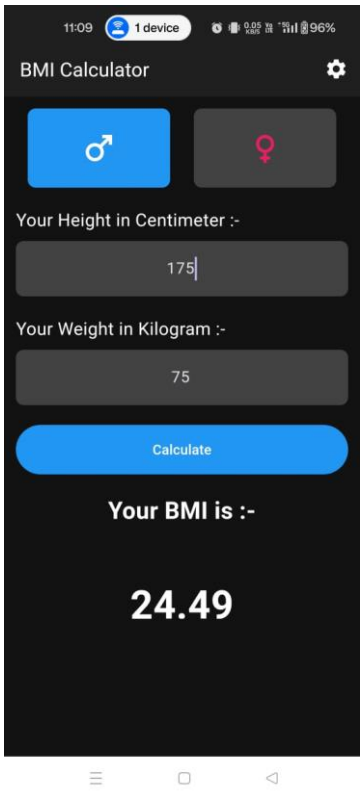
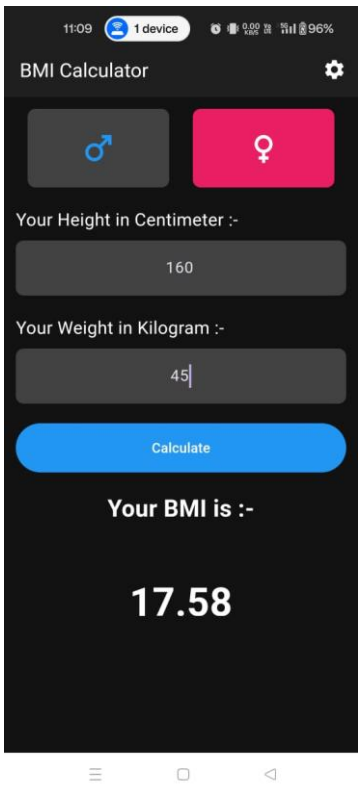
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** To design Flutter UI by including common widgets, Design BMI Calculator as per given screenshot	
Experiment No:- 03	**Date:-** 26-09-2024	**Enrollment No:-** 92200133030

Output:-







Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** To create an interactive Form using the form widget, Create a Loan Calculator(EMI based on simple as well as compound.	
Experiment No:- 04	**Date:-** 01-10-2024	**Enrollment No:-** 92200133030

Objective:- To create an interactive Form using the form widget, Create a Loan Calculator(EMI based on simple as well as compound.

Code :-

```
import 'package:flutter/material.dart';
import 'dart:math'; // Import math package to use pow() for compound interest calculation

void main() {
  runApp(const MyApp());
}

// Main class for the app
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      theme: ThemeData(
        brightness: Brightness.light,
        primaryColor: Colors.blue, // Light theme with blue as primary color
      ),
      darkTheme: ThemeData(
        brightness: Brightness.dark,
        scaffoldBackgroundColor:
            const Color(0xFF121212), // Dark theme background color
        appBarTheme: const AppBarTheme(
          backgroundColor: Color(0xFF1F1F1F),
          iconTheme: IconThemeData(color: Colors.white),
          titleTextStyle: TextStyle(color: Colors.white, fontSize: 20),
        ),
        textButtonTheme: TextButtonThemeData(
          style: TextButton.styleFrom(
            foregroundColor: Colors.white,
            backgroundColor: Colors.blue,
          ),
        ),
        inputDecorationTheme: InputDecorationTheme(
          filled: true,
          fillColor: Colors.grey[800],
          border: OutlineInputBorder(
            borderRadius: BorderRadius.circular(8.0),
            borderSide: BorderSide.none,
          ),
          hintStyle: const TextStyle(color: Colors.grey),
        ),
        textTheme: const TextTheme(
          bodyMedium: TextStyle(color: Colors.white),
          titleLarge: TextStyle(color: Colors.white),
```

```
        ),
      ),
      themeMode: ThemeMode.dark, // Set dark mode by default
      home: const EMICalculator(), // Set the EMI calculator screen as the home
      debugShowCheckedModeBanner: false, // Hide the debug banner
    );
  }
}

// Stateful widget for EMI Calculator
class EMICalculator extends StatefulWidget {
  const EMICalculator({super.key});

  @override
  State<EMICalculator> createState() => _EMICalculatorState();
}

class _EMICalculatorState extends State<EMICalculator> {
  int currentindex = 0; // 0 for Simple Interest, 1 for Compound Interest
  String result = ""; // To hold the calculated EMI result
  double principal = 0.0; // Principal amount
  double rate = 0.0; // Annual interest rate
  double tenure = 0.0; // Loan tenure in years
  double totalPayment = 0.0; // Total payment calculated
  double totalInterest = 0.0; // Total interest calculated

  // Controllers for TextField inputs
  TextEditingController principalController = TextEditingController();
  TextEditingController rateController = TextEditingController();
  TextEditingController tenureController = TextEditingController();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("EMI Calculator"), // AppBar title
        backgroundColor: Theme.of(context)
            .appBarTheme
            .backgroundColor, // AppBar color based on theme
        actions: [
          IconButton(
            onPressed: () {},
            icon:
                const Icon(Icons.settings), // Settings icon (currently unused)
          ),
        ],
      ),
      body: SingleChildScrollView(
        // Enables scrolling for the screen
        child: Padding(
          padding: const EdgeInsets.all(12.0), // Screen padding
          child: Column(
```

```
mainAxisAlignment: MainAxisAlignment.start,
crossAxisAlignment: CrossAxisAlignment.start,
children: [
  Row(
    children: [
      // Radio buttons for selecting Simple/Compound interest
      radioButton("Simple Interest", Colors.blue, 0),
      radioButton("Compound Interest", Colors.pink, 1),
    ],
  ),
  const SizedBox(height: 20.0),
  const Text(
    "Loan Amount (Principal) :- ",
    style: TextStyle(fontSize: 18.0),
  ),
  const SizedBox(height: 8.0),
  // TextField for entering principal amount
  TextField(
    controller: principalController,
    keyboardType: TextInputType.number,
    textAlign: TextAlign.center,
    decoration: InputDecoration(
      hintText: "Enter Principal Amount",
      filled: true,
      fillColor: Theme.of(context)
          .inputDecorationTheme
          .fillColor, // TextField styling based on theme
      border: OutlineInputBorder(
        borderRadius: BorderRadius.circular(8.0),
        borderSide: BorderSide.none,
      ),
    ),
  ),
  const SizedBox(height: 20.0),
  const Text(
    "Annual Interest Rate (%) :- ",
    style: TextStyle(fontSize: 18.0),
  ),
  const SizedBox(height: 8.0),
  // TextField for entering interest rate
  TextField(
    controller: rateController,
    keyboardType: TextInputType.number,
    textAlign: TextAlign.center,
    decoration: InputDecoration(
      hintText: "Enter Interest Rate",
      filled: true,
      fillColor: Theme.of(context).inputDecorationTheme.fillColor,
      border: OutlineInputBorder(
        borderRadius: BorderRadius.circular(8.0),
        borderSide: BorderSide.none,
      ),
```

```
    ),
  ),
  const SizedBox(height: 20.0),
  const Text(
    "Loan Tenure (in Years) :- ",
    style: TextStyle(fontSize: 18.0),
  ),
  const SizedBox(height: 8.0),
  // TextField for entering tenure in years
  TextField(
    controller: tenureController,
    keyboardType: TextInputType.number,
    textAlign: TextAlign.center,
    decoration: InputDecoration(
      hintText: "Enter Loan Tenure",
      filled: true,
      fillColor: Theme.of(context).inputDecorationTheme.fillColor,
      border: OutlineInputBorder(
        borderRadius: BorderRadius.circular(8.0),
        borderSide: BorderSide.none,
      ),
    ),
  ),
  const SizedBox(height: 20.0),
  SizedBox(
    width: double.infinity, // Button fills full width
    height: 50.0, // Button height
    child: TextButton(
      onPressed: () {
        // Parsing values from TextField inputs
        setState(() {
          principal =
              double.tryParse(principalController.text) ?? 0.0;
          rate = double.tryParse(rateController.text) ?? 0.0;
          tenure = double.tryParse(tenureController.text) ?? 0.0;
        });
        // Call calculation method based on selected interest type
        if (currentindex == 0) {
          calculateEMIWithSimpleInterest(principal, rate, tenure);
        } else {
          calculateEMIWithCompoundInterest(principal, rate, tenure);
        }
      },
      child: const Text(
        "Calculate EMI",
        style: TextStyle(color: Colors.white),
      ),
    ),
  ),
  const SizedBox(height: 20.0),
  const Center(
    child: Text(
```

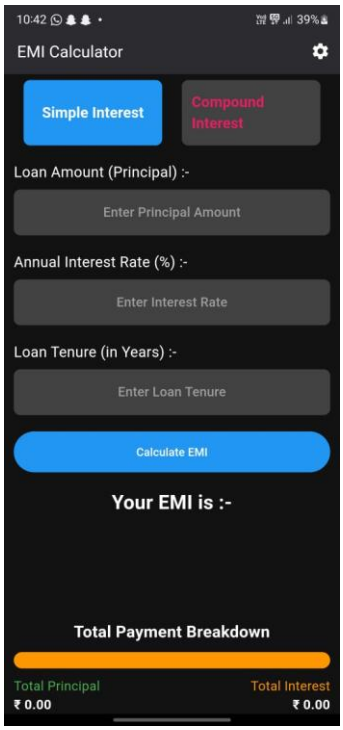
```
      "Your EMI is :- ",
      textAlign: TextAlign.center,
      style: TextStyle(
        fontSize: 24.0,
        fontWeight: FontWeight.bold,
      ),
    ),
  ),
  const SizedBox(height: 50.0),
  Center(
    child: Text(
      result, // Display calculated EMI
      textAlign: TextAlign.center,
      style: const TextStyle(
        fontSize: 40.0,
        fontWeight: FontWeight.bold,
      ),
    ),
  ),
  const SizedBox(height: 20.0),
  const Center(
    child: Text(
      "Total Payment Breakdown",
      textAlign: TextAlign.center,
      style: TextStyle(
        fontSize: 20.0,
        fontWeight: FontWeight.bold,
      ),
    ),
  ),
  const SizedBox(height: 10.0),
  // Linear progress bar indicating total payment breakdown
  Container(
    height: 20.0,
    decoration: BoxDecoration(
      color: Colors.orange, // Background color of the progress bar
      borderRadius: BorderRadius.circular(10.0),
    ),
    child: ClipRRect(
      borderRadius: BorderRadius.circular(10.0),
      child: LinearProgressIndicator(
        value: (principal > 0 && totalPayment > 0)
            ? principal / totalPayment
            : 0.0,
        backgroundColor: Colors.transparent,
        valueColor:
            const AlwaysStoppedAnimation<Color>(Colors.green),
      ),
    ),
  ),
  const SizedBox(height: 10.0),
  // Displaying total principal and interest
```

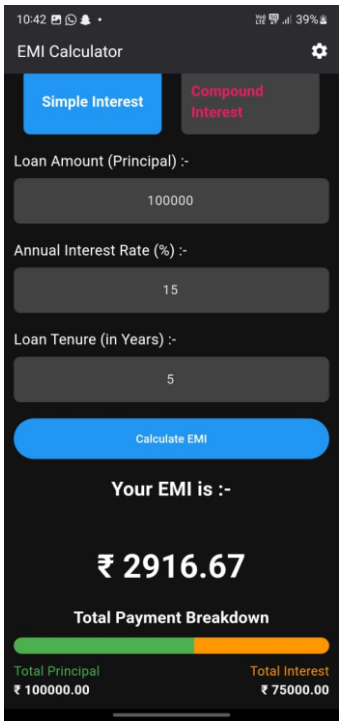
```
            Row(
              mainAxisAlignment: MainAxisAlignment.spaceBetween,
              children: [
                Column(
                  crossAxisAlignment: CrossAxisAlignment.start,
                  children: [
                    Text(
                      "Total Principal",
                      style: TextStyle(fontSize: 16.0, color: Colors.green),
                    ),
                    Text(
                      "₹ ${principal.toStringAsFixed(2)}",
                      style: const TextStyle(
                        fontSize: 16.0,
                        fontWeight: FontWeight.bold,
                      ),
                    ),
                  ],
                ),
                Column(
                  crossAxisAlignment: CrossAxisAlignment.end,
                  children: [
                    Text(
                      "Total Interest",
                      style: TextStyle(fontSize: 16.0, color: Colors.orange),
                    ),
                    Text(
                      "₹ ${totalInterest.toStringAsFixed(2)}",
                      style: const TextStyle(
                        fontSize: 16.0,
                        fontWeight: FontWeight.bold,
                      ),
                    ),
                  ],
                ),
              ],
            ),
          ],
        ),
      ),
    ),
  );
}

// Method for radio button widget
Widget radioButton(String value, Color color, int index) {
  return Expanded(
    child: Container(
      margin: const EdgeInsets.symmetric(horizontal: 12.0),
      height: 50.0,
      child: TextButton(
        onPressed: () {
```

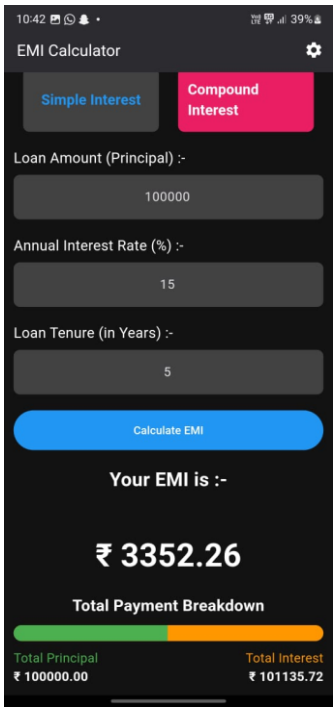
```
            setState(() {
              currentindex = index; // Update index when radio button is pressed
            });
          },
          style: TextButton.styleFrom(
            backgroundColor: currentindex == index ? color : Colors.grey[200],
          ),
          child: Text(
            value,
            style: TextStyle(
              color: currentindex == index ? Colors.white : color,
            ),
          ),
        ),
      ),
    ),
  );
}

// Method to calculate EMI with Simple Interest
void calculateEMIWithSimpleInterest(
    double principal, double rate, double tenure) {
  setState(() {
    double interest =
        (principal * rate * tenure) / 100; // Simple interest formula
    totalPayment = principal + interest; // Total payment
    totalInterest = interest; // Total interest
    result = (totalPayment / (tenure * 12))
        .toStringAsFixed(2); // EMI calculation for months
  });
}

// Method to calculate EMI with Compound Interest
void calculateEMIWithCompoundInterest(
    double principal, double rate, double tenure) {
  setState(() {
    double monthlyRate = rate / (12 * 100); // Monthly interest rate
    int n = (tenure * 12).toInt(); // Number of months (n)
    totalPayment =
        principal * pow(1 + monthlyRate, n); // Compound interest formula
    totalInterest = totalPayment - principal; // Total interest
    result =
        (totalPayment / n).toStringAsFixed(2); // EMI calculation for months
  });
}
}
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** To create an interactive Form using the form widget, Create a Loan Calculator(EMI based on simple as well as compound.
Experiment No:- 04	**Date:-** 01-10-2024 **Enrollment No:-** 92200133030

Output:-







Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** To design a layout of the Flutter App using layout widgets. Create a flutter app to manage Todos with the add , remove , complete and list of todo.	
Experiment No:- 05	**Date:-** 03-10-2024	**Enrollment No:-** 92200133030

Objective:- To design a layout of the Flutter App using layout widgets. Create a flutter app to manage Todos with the add , remove , complete and list of todo.

Code :-

- **lib**
  - **constants**
    - **colors.dart**

```
import 'package:flutter/material.dart';

// Define a constant color for red, typically used for errors or alerts
const Color tdRed = Color(0xFFDA4040);

// Define a constant color for blue, can be used for primary buttons or highlights
const Color tdBlue = Color(0xFF5F52EE);

// Define a constant color for black, useful for text or dark backgrounds
const Color tdBlack = Color(0xFF3A3A3A);

// Define a constant color for grey, often used for disabled elements or secondary
text
const Color tdGrey = Color(0xFF717171);

// Define a constant background color, providing a light theme to the app
const Color tdBGColor = Color(0xFFEEEFF5);
```

  - **controller**
    - **todo_item.dart**

```
import 'package:flutter/material.dart';

import '../model/todo.dart'; // Importing the ToDo model.
import '../constants/colors.dart'; // Importing custom color constants.

class ToDoItem extends StatelessWidget {
  final ToDo todo; // The ToDo item to be displayed.
  // Callback function when the ToDo item is changed (completed or not).
  // ignore: prefer_typing_uninitialized_variables
  final onToDoChanged;
  // Callback function when the delete icon is pressed.
  // ignore: prefer_typing_uninitialized_variables
  final onDeleteItem;

  const ToDoItem({
    super.key,
```

```
  required this.todo, // Required parameter for the ToDo item.
  required this.onToDoChanged, // Required callback for item change.
  required this.onDeleteItem, // Required callback for item deletion.
});

@override
Widget build(BuildContext context) {
  return Container(
    margin: const EdgeInsets.only(bottom: 20), // Space between items.
    child: ListTile(
      onTap: () {
        // This triggers when the ToDo item is tapped.
        // print('Clicked on Todo Item.'); // Uncomment for debugging.
        onToDoChanged(todo); // Call the callback to update ToDo status.
      },
      shape: RoundedRectangleBorder(
        borderRadius:
            BorderRadius.circular(20), // Rounded corners for the tile.
      ),
      contentPadding: const EdgeInsets.symmetric(
          horizontal: 20, vertical: 5), // Padding inside the tile.
      tileColor: Colors.white, // Background color of the tile.
      leading: Icon(
        // Icon showing whether the ToDo is completed or not.
        todo.isDone ? Icons.check_box : Icons.check_box_outline_blank,
        color: tdBlue, // Color for the checkbox icon.
      ),
      title: Text(
        todo.todoText!, // Display the text of the ToDo item.
        style: TextStyle(
          fontSize: 16, // Font size for the ToDo text.
          color: tdBlack, // Text color.
          decoration: todo.isDone
              ? TextDecoration.lineThrough
              : null, // Strikethrough if done.
        ),
      ),
      trailing: Container(
        padding: const EdgeInsets.all(
            0), // No padding inside the trailing container.
        margin: const EdgeInsets.symmetric(
            vertical: 12), // Vertical margin for the container.
        height: 35, // Fixed height for the delete button.
        width: 35, // Fixed width for the delete button.
        decoration: BoxDecoration(
          color: tdRed, // Background color for the delete button.
          borderRadius: BorderRadius.circular(
              5), // Rounded corners for the delete button.
        ),
        child: IconButton(
          color: Colors.white, // Icon color.
          iconSize: 18, // Size of the delete icon.
```

```
                icon: const Icon(Icons.delete), // Delete icon.
                onPressed: () {
                  // This triggers when the delete icon is pressed.
                  // print('Clicked on delete icon'); // Uncomment for debugging.
                  onDeleteItem(
                      todo.id); // Call the callback to delete the ToDo item.
                },
              ),
            ),
          ),
        ),
      );
    }
  }
```

- **model**
  - **todo.dart**

```
class ToDo {
  // Unique identifier for the ToDo item
  String? id;

  // The text description of the ToDo item
  String? todoText;

  // Flag to indicate whether the ToDo item is completed or not
  bool isDone;

  // Constructor for the ToDo class, with 'id' and 'todoText' being required
  // 'isDone' is optional and defaults to false
  ToDo({
    required this.id,
    required this.todoText,
    this.isDone = false,
  });

  // Static method to return an empty list of ToDo items
  static List<ToDo> todoList() {
    return []; // Returns an empty list of ToDo objects
  }
}
```

- **view**
  - **home.dart**

```
import 'package:flutter/material.dart';
import 'package:flutter/services.dart';
import '../model/todo.dart'; // Importing the ToDo model
import '../constants/colors.dart'; // Importing color constants
import '../controller/todo_item.dart'; // Importing the ToDoItem widget

// The main Home widget, a StatefulWidget to manage state
class Home extends StatefulWidget {
  const Home({super.key});

  @override
  State<Home> createState() => _HomeState();
}

// State class for Home
class _HomeState extends State<Home> {
  // List to hold all the ToDo items
  final todosList =
      ToDo.todoList(); // Initializing the list with predefined items
  List<ToDo> _foundToDo = []; // List to hold filtered ToDo items
  final _todoController =
      TextEditingController(); // Controller for the TextField

  @override
  void initState() {
    _foundToDo = todosList; // Initialize _foundToDo with all todos
    super.initState();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: tdBGColor, // Setting background color
      appBar: _buildAppBar(), // Building the app bar
      body: Stack(
        children: [
          Container(
            padding: const EdgeInsets.symmetric(
              horizontal: 20,
              vertical: 15,
            ),
            child: Column(
              children: [
                searchBox(), // Search box for filtering todos
                Expanded(
                  child: ListView(
                    children: [
                      Container(
```

```
                margin: const EdgeInsets.only(
                  top: 50,
                  bottom: 20,
                ),
                child: const Text(
                  'All ToDos', // Title for the ToDo list
                  style: TextStyle(
                    fontSize: 30,
                    fontWeight: FontWeight.w500,
                  ),
                ),
              ),
              // Displaying each ToDo item in reverse order
              for (ToDo todoo in _foundToDo.reversed)
                ToDoItem(
                  todo: todoo, // Passing the current ToDo item
                  onToDoChanged:
                      _handleToDoChange, // Callback for when ToDo is
changed
                  onDeleteItem:
                      _deleteToDoItem, // Callback for deleting a ToDo
item
                ),
            ],
          ),
        )
      ],
    ),
  ),
  // Bottom input field and button for adding new ToDo items
  Align(
    alignment: Alignment.bottomCenter,
    child: Row(children: [
      Expanded(
        child: Container(
          margin: const EdgeInsets.only(
            bottom: 20,
            right: 20,
            left: 20,
          ),
          padding: const EdgeInsets.symmetric(
            horizontal: 20,
            vertical: 5,
          ),
          decoration: BoxDecoration(
            color: Colors.white, // White background for input field
            boxShadow: const [
              BoxShadow(
                color: Colors.grey, // Shadow effect
                offset: Offset(0.0, 0.0),
                blurRadius: 10.0,
                spreadRadius: 0.0,
```

```
                ),
              ],
              borderRadius: BorderRadius.circular(10), // Rounded corners
            ),
            child: TextField(
              controller: _todoController, // TextField controller
              decoration: const InputDecoration(
                  hintText: 'Add a new todo item', // Placeholder text
                  border: InputBorder.none), // No border
            ),
          ),
        ),
        Container(
          margin: const EdgeInsets.only(
            bottom: 20,
            right: 20,
          ),
          child: ElevatedButton(
            onPressed: () {
              _addToDoItem(_todoController.text); // Adding new ToDo item
            },
            style: ElevatedButton.styleFrom(
              backgroundColor: tdBlue, // Button background color
              minimumSize: const Size(60, 60), // Button size
              elevation: 10, // Button shadow
            ),
            child: const Text(
              '+', // Button text
              style: TextStyle(
                  fontSize: 40, color: Colors.white), // Text style
            ),
          ),
        ),
      ]),
    ),
  ],
),
);
}

// Handle change of ToDo item status (completed or not)
void _handleToDoChange(ToDo todo) {
  setState(() {
    todo.isDone = !todo.isDone; // Toggle the isDone status
  });
}

// Delete ToDo item from the list by its ID
void _deleteToDoItem(String id) {
  setState(() {
    todosList
        .removeWhere((item) => item.id == id); // Remove item from the list
```

```
  });
}

// Add new ToDo item to the list
void _addToDoItem(String toDo) {
  setState(() {
    todosList.add(ToDo(
      id: DateTime.now()
          .millisecondsSinceEpoch
          .toString(), // Unique ID based on timestamp
      todoText: toDo, // ToDo text
    ));
  });
  _todoController.clear(); // Clear the text field after adding
}

// Filter ToDo items based on search input
void _runFilter(String enteredKeyword) {
  List<ToDo> results = [];
  if (enteredKeyword.isEmpty) {
    results = todosList; // If search input is empty, show all todos
  } else {
    results = todosList
        .where((item) => item.todoText!.toLowerCase().contains(
            enteredKeyword.toLowerCase())) // Filter based on search input
        .toList();
  }

  setState(() {
    _foundToDo = results; // Update found ToDo list
  });
}

// Widget for the search box
Widget searchBox() {
  return Container(
    padding: const EdgeInsets.symmetric(horizontal: 15),
    decoration: BoxDecoration(
      color: Colors.white, // White background for the search box
      borderRadius: BorderRadius.circular(20), // Rounded corners
    ),
    child: TextField(
      onChanged: (value) => _runFilter(value), // Run filter on text change
      decoration: const InputDecoration(
        contentPadding: EdgeInsets.all(0), // Padding for the input
        prefixIcon: Icon(
          Icons.search, // Search icon
          color: tdBlack,
          size: 20,
        ),
        prefixIconConstraints: BoxConstraints(
          maxHeight: 20,
```

```
                minWidth: 25,
              ),
              border: InputBorder.none, // No border for the input field
              hintText: 'Search', // Placeholder text
              hintStyle: TextStyle(color: tdGrey), // Hint text style
            ),
          ),
        );
      }

      // Widget to build the app bar
      AppBar _buildAppBar() {
        return AppBar(
          backgroundColor: tdBGColor, // App bar background color
          elevation: 0, // No shadow
          title: const Row(
              mainAxisAlignment: MainAxisAlignment.spaceBetween,
              children: [
                Icon(
                  Icons.menu, // Menu icon
                  color: tdBlack,
                  size: 30,
                ),
              ]),
        );
      }
    }

    // Main application widget
    class MyApp extends StatelessWidget {
      const MyApp({super.key});

      // This widget is the root of your application.
      @override
      Widget build(BuildContext context) {
        SystemChrome.setSystemUIOverlayStyle(const SystemUiOverlayStyle(
            statusBarColor: Colors.transparent)); // Setting the status bar color
        return const MaterialApp(
          debugShowCheckedModeBanner: false, // Disable debug banner
          title: 'ToDo App', // Application title
          home: Home(), // Home widget as the main screen
        );
      }
    }
```

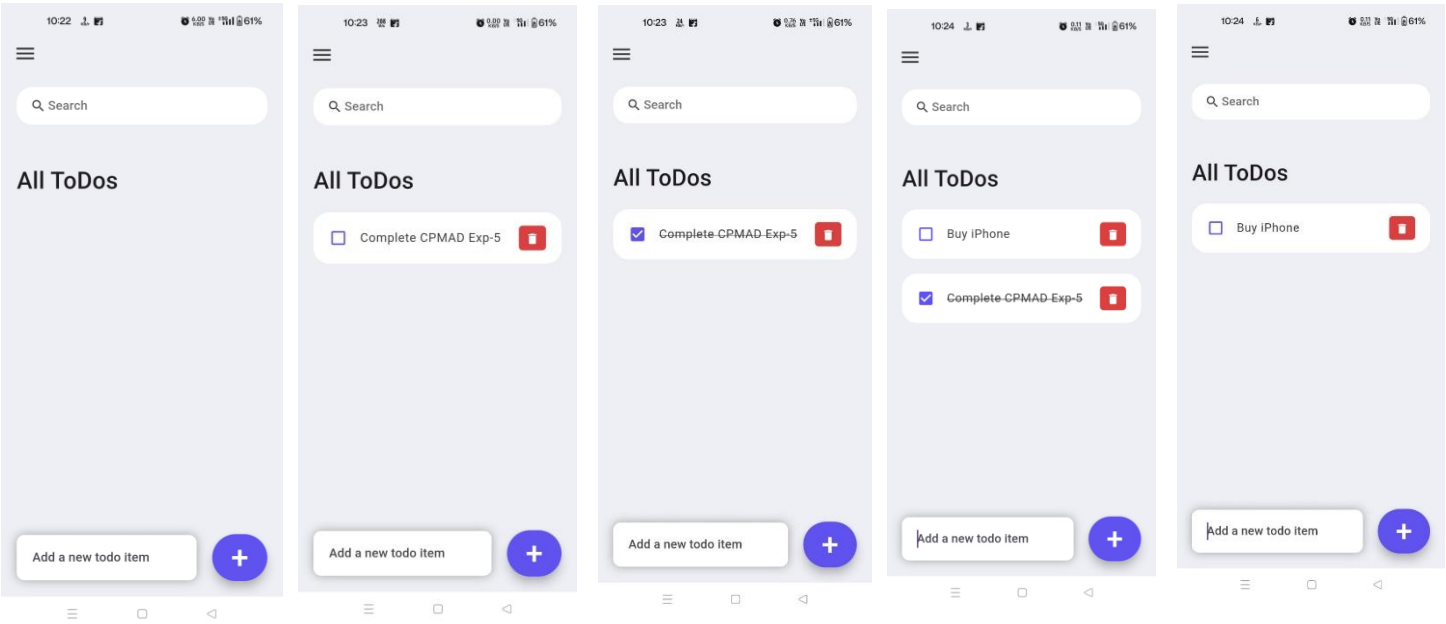
- **main.dart**

```
import 'package:flutter/material.dart'; // Importing Flutter material package for
UI components
import 'package:flutter/services.dart'; // Importing Flutter services package for
system UI control
import './view/home.dart'; // Importing the Home widget from the view directory
```

```
void main() {
  runApp(
      const MyApp()); // Running the MyApp widget as the root of the application
}

class MyApp extends StatelessWidget {
  const MyApp({super.key}); // Constructor for MyApp with a key parameter

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    // Setting the system UI overlay style to make the status bar transparent
    SystemChrome.setSystemUIOverlayStyle(
        const SystemUiOverlayStyle(statusBarColor: Colors.transparent));

    // Building the MaterialApp widget which serves as the main app structure
    return const MaterialApp(
      debugShowCheckedModeBanner: false, // Hiding the debug banner in the app
      title: 'ToDo App', // Setting the title of the application
      home: Home(), // Specifying the Home widget as the starting screen
    );
  }
}
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** To design a layout of the Flutter App using layout widgets. Create a flutter app to manage Todos with the add , remove , complete and list of todo.	
Experiment No:- 05	**Date:-** 03-10-2024	**Enrollment No:-** 92200133030

Output:-



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** Daily Expense app using getx.	
Experiment No:- 06	**Date:-** 18-10-2024	**Enrollment No:-** 92200133030

Objective:- Daily Expense app using getx.

Code :-

- **lib**
  - **controller**
    - **expense_controller.dart**

```
import 'package:expense_manager/model/expense_data.dart';
import 'package:flutter/material.dart';
import 'package:get/get.dart';
import 'package:intl/intl.dart';

class ExpenseController extends GetxController {
  var selected = "All".obs;
  var curBottomNavIndex = 0.obs;

  var income = 0.0.obs;
  var spent = 0.0.obs;

  RxList<ExpenseData> expenses = <ExpenseData>[].obs;

  var category = "Food".obs;
  var paymentType = "Cash".obs;

  TextEditingController amountCntrl = TextEditingController();

  double amount = 0.0;

  List<List<dynamic>> iconList = [
    [Icons.home, "Home"],
    [Icons.swap_horiz, "Transfer"],
    [Icons.account_balance_wallet_outlined, "Wallet"],
    [Icons.account_circle_outlined, "Profile"],
  ];

  void saveData() {
    final newExpense = ExpenseData(
      amount: double.tryParse(amountCntrl.text) ?? 0,
      category: category.value,
      paymentType: paymentType.value,
      timestamp: DateFormat('MMM dd, yyyy').format(DateTime.now()),
    );

    expenses.add(newExpense);
    if (category.value == "Salary") {
      income.value += newExpense.amount;
    } else {
      spent.value += newExpense.amount;
```

```
    }
  }
}
```

- **model**
  - **expense_data.dart**

```
class ExpenseData {
  double amount;
  String category;
  String paymentType;
  String timestamp;

  ExpenseData({
    required this.amount,
    required this.category,
    required this.paymentType,
    required this.timestamp,
  });

  Map<String, dynamic> toMap() {
    return {
      'amount': amount,
      'category': category,
      'paymentType': paymentType,
      'timestamp': timestamp,
    };
  }

  factory ExpenseData.fromMap(Map<String, dynamic> map) {
    return ExpenseData(
      amount: map['amount'],
      category: map['category'],
      paymentType: map['paymentType'],
      timestamp: map['timestamp'],
    );
  }
}
```

- **screen**
  - **home_page.dart**

```
import 'package:animated_bottom_navigation_bar/animated_bottom_navigation_bar.dart';
import 'package:charts_flutter/flutter.dart' as charts;
import 'package:expense_manager/controller/expense_contoller.dart';
import 'package:expense_manager/screen/add_expense.dart';
import 'package:flutter/material.dart';
import 'package:get/get.dart';

class HomePage extends StatelessWidget {
  const HomePage({super.key});
  static ExpenseController controller = Get.put(ExpenseController());

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Container(
        color: const Color(0xFFb6d8ae),
        child: SafeArea(
          child: Container(
            decoration: const BoxDecoration(
              color: Colors.white,
              borderRadius: BorderRadius.only(
                topLeft: Radius.circular(30),
                topRight: Radius.circular(30),
              ),
            ),
            padding: const EdgeInsets.fromLTRB(20, 20, 20, 0),
            child: Column(
              children: [
                Row(
                  children: [
                    const Expanded(
                      child: Column(
                        crossAxisAlignment: CrossAxisAlignment.start,
                        children: [
                          Text(
                            "Hello,",
                            style: TextStyle(fontSize: 24),
                          ),
                          Text(
                            "Aryan Langhanoja",
                            style: TextStyle(
                                fontSize: 24, fontWeight: FontWeight.bold),
                          )
                        ],
                      ),
                    ),
                    Container(
                      decoration: BoxDecoration(
```

```
            border: Border.all(color: Colors.grey[300]!),
            borderRadius: BorderRadius.circular(32),
          ),
          child: IconButton(
            onPressed: () {},
            icon: const Icon(Icons.notifications_none),
          ),
        ),
      ],
    ),
    Obx(
      () => SingleChildScrollView(
        scrollDirection: Axis.horizontal,
        child: Row(
          children: [
            outlinedButton("All", () {
              controller.selected.value = "All";
            }),
            outlinedButton("Daily", () {
              controller.selected.value = "Daily";
            }),
            outlinedButton("Weekly", () {
              controller.selected.value = "Weekly";
            }),
            outlinedButton("Monthly", () {
              controller.selected.value = "Monthly";
            }),
            outlinedButton("Yearly", () {
              controller.selected.value = "Yearly";
            }),
          ],
        ),
      ),
    ),
    Container(
      margin: const EdgeInsets.only(top: 20, bottom: 15),
      decoration: BoxDecoration(
        borderRadius: BorderRadius.circular(15),
        color: Colors.grey[100],
      ),
      child: Row(
        mainAxisAlignment: MainAxisAlignment.spaceAround,
        children: [
          incomeSpentSection(),
          pieChartSection(),
        ],
      ),
    ),
    Row(
      children: [
        const Expanded(
          child: Text(
```

```
                "Recent transactions",
                style: TextStyle(color: Colors.grey, fontSize: 20),
              ),
            ),
            OutlinedButton(
              onPressed: () {},
              style: OutlinedButton.styleFrom(
                padding: const EdgeInsets.fromLTRB(15, 0, 5, 0),
                shape: RoundedRectangleBorder(
                  borderRadius: BorderRadius.circular(32),
                ),
                side: BorderSide(color: Colors.grey[300]!, width: 1.5),
              ),
              child: const Row(
                children: [
                  Text(
                    "See All",
                    style: TextStyle(color: Colors.grey),
                  ),
                  Icon(
                    Icons.chevron_right,
                    color: Colors.grey,
                  )
                ],
              ),
            ),
          ],
        ),
        Expanded(
          child: Obx(
            () => ListView.builder(
              itemCount: controller.expenses.length,
              itemBuilder: (context, index) {
                final expense = controller.expenses[index];
                return expenseCard(
                  expense.amount,
                  expense.category,
                  expense.paymentType,
                  expense.timestamp,
                );
              },
            ),
          ),
        ),
      ],
    ),
  ),
),
floatingActionButton: FloatingActionButton(
  onPressed: () {
    Get.to(() => AddExpense());
```

```
        },
        backgroundColor: const Color(0xFFb7d9ae),
        child: Icon(
          Icons.add,
          color: Colors.grey[800],
        ),
      ),
      floatingActionButtonLocation: FloatingActionButtonLocation.centerDocked,
      bottomNavigationBar: Obx(
        () => AnimatedBottomNavigationBar.builder(
          backgroundColor: Colors.white,
          itemCount: controller.iconList.length,
          activeIndex: controller.curBottomNavIndex.value,
          onTap: (index) => controller.curBottomNavIndex.value = index,
          splashSpeedInMilliseconds: 300,
          notchSmoothness: NotchSmoothness.defaultEdge,
          gapLocation: GapLocation.center,
          tabBuilder: (int index, bool isActive) {
            return Column(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [
                Icon(
                  controller.iconList[index][0],
                  size: 24,
                  color: isActive ? const Color(0xFFb7d9ae) : Colors.grey[800],
                ),
                const SizedBox(height: 4),
                Padding(
                  padding: const EdgeInsets.symmetric(horizontal: 8),
                  child: Text(
                    controller.iconList[index][1],
                    style: TextStyle(color: Colors.grey[800]),
                  ),
                ),
              ],
            );
          },
        ),
      ),
    );
  }

  Widget outlinedButton(String label, VoidCallback onPressed) {
    return Padding(
      padding: const EdgeInsets.fromLTRB(0, 0, 10, 0),
      child: OutlinedButton(
        onPressed: onPressed,
        style: OutlinedButton.styleFrom(
          padding: const EdgeInsets.symmetric(horizontal: 20, vertical: 12),
          shape: RoundedRectangleBorder(
            borderRadius: BorderRadius.circular(15),
          ),
```

```
            side: BorderSide(
              color: controller.selected.value == label
                  ? const Color(0xFFb6d8ae)
                  : Colors.grey[300]!,
              width: 1.5,
            ),
          ),
          child: Text(
            label,
            style: TextStyle(
              color:
                  controller.selected.value != label ? Colors.grey : Colors.black,
            ),
          ),
        ),
      );
    }

    Widget incomeSpentSection() {
      return Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          labelAndValue("Income", controller.income, const Color(0xFFb6d8ae)),
          const SizedBox(height: 20),
          labelAndValue("Spent", controller.spent, const Color(0xFFfeb9ab)),
        ],
      );
    }

    Widget labelAndValue(String label, RxDouble value, Color color) {
      return Column(
        children: [
          Row(
            children: [
              Container(
                width: 5,
                height: 15,
                decoration: BoxDecoration(
                  borderRadius: BorderRadius.circular(3),
                  color: color,
                ),
              ),
              Text(" $label"),
            ],
          ),
          Obx(
            () => Text(
              "₹${value.value % 1 == 0.0 ? value.value.toInt() :
    value.value.toStringAsFixed(2)}",
              style: const TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
            ),
          ),
```

```
    ],
  );
}

Widget pieChartSection() {
  return Obx(
    () => SizedBox(
      width: 165,
      height: 165,
      child: charts.PieChart(
        [
          charts.Series<Expense, int>(
            id: 'Expense',
            domainFn: (Expense obj, _) => obj.label,
            measureFn: (Expense obj, _) => obj.value,
            data: [
              Expense(0, controller.income.value.toInt()),
              Expense(1, controller.spent.value.toInt()),
            ],
            colorFn: (datum, index) => [
              charts.ColorUtil.fromDartColor(const Color(0xFFb6d8ae)),
              charts.ColorUtil.fromDartColor(const Color(0xFFfeb9ab)),
            ][index!],
          ),
        ],
        animate: false,
        defaultRenderer: charts.ArcRendererConfig<int>(
          arcWidth: 30,
          startAngle: 31.4,
        ),
      ),
    ),
  );
}

Widget expenseCard(
    double amount, String category, String paymentType, String timestamp) {
  return Padding(
    padding: const EdgeInsets.fromLTRB(0, 0, 0, 15),
    child: Row(
      children: [
        Container(
          width: 48,
          height: 48,
          decoration: BoxDecoration(
            borderRadius: BorderRadius.circular(15),
            color: category == "Food"
                ? const Color(0xFFfaedd4)
                : category == "Salary"
                    ? const Color(0xFFe9f3e6)
                    : const Color(0xFFf1e6ff),
          ),
```

```
                child: Icon(
                  category == "Food"
                      ? Icons.emoji_food_beverage_outlined
                      : category == "Salary"
                          ? Icons.currency_rupee
                          : Icons.local_movies_outlined,
                  color: Colors.grey[800],
                ),
              ),
              const SizedBox(width: 15),
              Expanded(
                child: Column(
                  children: [
                    Row(
                      children: [
                        Expanded(
                          child: Text(
                            category,
                            style: const TextStyle(fontWeight: FontWeight.bold),
                          ),
                        ),
                        Text(
                          "₹${amount % 1 == 0 ? amount.toInt() : amount}",
                          style: const TextStyle(fontWeight: FontWeight.bold),
                        ),
                      ],
                    ),
                    Row(
                      children: [
                        Expanded(
                          child: Text(
                            paymentType,
                            style: const TextStyle(color: Colors.grey),
                          ),
                        ),
                        Text(
                          timestamp,
                          style: const TextStyle(color: Colors.grey),
                        ),
                      ],
                    ),
                  ],
                ),
              ),
            ],
          ),
        );
      }
    }
```

```
class Expense {
  final int label;
  final int value;

  Expense(this.label, this.value);
}
```

- ## add_expense.dart

```
import 'package:expense_manager/controller/expense_contoller.dart';
import 'package:flutter/material.dart';
import 'package:flutter/services.dart';
import 'package:get/get.dart';

class AddExpense extends StatelessWidget {
  AddExpense({super.key});
  final ExpenseController controller = Get.find<ExpenseController>();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: const Color(0xFFb6d8ae),
      body: SafeArea(
        child: Container(
          decoration: BoxDecoration(
            color: Colors.white,
            borderRadius: BorderRadius.circular(30),
          ),
          padding: const EdgeInsets.fromLTRB(20, 30, 20, 30),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Row(
                children: [
                  OutlinedButton(
                    onPressed: () {},
                    style: OutlinedButton.styleFrom(
                      padding: const EdgeInsets.fromLTRB(10, 14, 10, 14),
                      shape: RoundedRectangleBorder(
                        borderRadius: BorderRadius.circular(20),
                      ),
                      side: BorderSide(color: Colors.grey[300]!, width: 1.5),
                    ),
                    child: const Icon(
                      Icons.chevron_left,
                      color: Colors.grey,
                      size: 32,
                    ),
                  ),
                  SizedBox(width: MediaQuery.of(context).size.width * 0.09),
                  Text(
```

```
        "Add transaction",
        style: TextStyle(fontSize: 24, color: Colors.grey[800]),
      ),
    ],
  ),
  Expanded(
    child: ListView(
      children: [
        Container(
          decoration: BoxDecoration(
            borderRadius: BorderRadius.circular(15),
            color: Colors.grey[100],
          ),
          padding: const EdgeInsets.symmetric(
              horizontal: 20, vertical: 10),
          margin: const EdgeInsets.fromLTRB(0, 20, 0, 5),
          child: TextField(
            controller: controller.amountCntrl,
            inputFormatters: [
              FilteringTextInputFormatter.allow(
                  RegExp(r'^\d+\.?\d{0,}'),
                  replacementString: ""),
            ],
            onTapOutside: (event) =>
                FocusScope.of(context).unfocus(),
            keyboardType: TextInputType.number,
            decoration: InputDecoration(
              border: InputBorder.none,
              label: Text(
                "Amount",
                style: TextStyle(color: Colors.grey[700]),
              ),
              alignLabelWithHint: false,
              prefixText: "₹",
              prefixStyle:
                  TextStyle(color: Colors.grey[800], fontSize: 16),
            ),
          ),
        ),
        Container(
          width: double.infinity,
          decoration: BoxDecoration(
            borderRadius: BorderRadius.circular(15),
            color: Colors.grey[100],
          ),
          padding: const EdgeInsets.symmetric(
              horizontal: 20, vertical: 20),
          margin: const EdgeInsets.fromLTRB(0, 5, 0, 5),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Text(
```

```
                "Category",
                style: TextStyle(color: Colors.grey[700]),
              ),
              const SizedBox(height: 10),
              Obx(
                () => DropdownButton(
                  value: controller.category.value,
                  items: [
                    dropDownCard(Icons.emoji_food_beverage_outlined,
                        "Food", const Color(0xFFfaedd4)),
                    dropDownCard(Icons.currency_rupee, "Salary",
                        const Color(0xFFe9f3e6)),
                    dropDownCard(Icons.local_movies_outlined,
                        "Entertainment", const Color(0xFFf1e6ff)),
                  ],
                  icon: const Icon(Icons.keyboard_arrow_down),
                  isExpanded: true,
                  underline: const SizedBox(),
                  onChanged: (value) {
                    controller.category.value = value!;
                    // setState(() => controller.category = value!);
                  },
                ),
              ),
            ],
          ),
        ),
        const SizedBox(height: 20),
        Text(
          "Payment Type",
          style: TextStyle(
            fontSize: 24,
            fontWeight: FontWeight.bold,
            color: Colors.grey[800],
          ),
        ),
        const SizedBox(height: 20),
        Obx(
          () => Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              radioButtonCard("Cash", 130),
              radioButtonCard("Credit/Debit Card", 220),
              radioButtonCard("Check", 140),
            ],
          ),
        ),
      ],
    ),
  ),
  Row(
    children: [
```

```
                Expanded(
                  child: OutlinedButton(
                    onPressed: () {},
                    style: OutlinedButton.styleFrom(
                      padding: const EdgeInsets.symmetric(vertical: 20),
                      shape: RoundedRectangleBorder(
                        borderRadius: BorderRadius.circular(20),
                      ),
                    ),
                    child: Text(
                      "Draft",
                      style: TextStyle(
                          fontSize: 18,
                          fontWeight: FontWeight.w600,
                          color: Colors.grey[800]),
                    ),
                  ),
                ),
                const SizedBox(width: 20),
                Expanded(
                  child: OutlinedButton(
                    onPressed: () {
                      controller.saveData();
                      Get.back();
                    },
                    style: OutlinedButton.styleFrom(
                      backgroundColor: const Color(0xFFb6d8ae),
                      padding: const EdgeInsets.symmetric(vertical: 20),
                      shape: RoundedRectangleBorder(
                        borderRadius: BorderRadius.circular(20),
                      ),
                    ),
                    child: Text(
                      "Add",
                      style: TextStyle(
                          fontSize: 18,
                          fontWeight: FontWeight.w600,
                          color: Colors.grey[800]),
                    ),
                  ),
                ),
              ],
            ),
          ],
        ),
      ),
    ),
  );
}

DropdownMenuItem dropDownCard(IconData icon, String label, Color backColor) {
  return DropdownMenuItem(
```

```
      value: label,
      child: Row(
        children: [
          Container(
            width: 42,
            height: 42,
            decoration: BoxDecoration(
              borderRadius: BorderRadius.circular(15),
              color: backColor,
            ),
            child: Icon(
              icon,
              color: Colors.grey[800],
              size: 20,
            ),
          ),
          const SizedBox(width: 15),
          Text(
            label,
            style: TextStyle(
                fontSize: 18,
                color: Colors.grey[800],
                fontWeight: FontWeight.bold),
          ),
        ],
      ),
    );
  }

  Widget radioButtonCard(String label, double width) {
    return Container(
      decoration: BoxDecoration(
        border: Border.all(
          color: controller.paymentType.value == label
              ? const Color(0xFFb6d8ae)
              : Colors.grey,
          width: 1.5,
        ),
        borderRadius: BorderRadius.circular(20),
      ),
      width: width,
      margin: const EdgeInsets.fromLTRB(0, 4, 0, 4),
      child: RadioListTile<String>(
        title: Text(
          label,
          style: TextStyle(
              color: controller.paymentType.value == label
                  ? Colors.grey[800]
                  : Colors.grey[600],
              fontSize: 16,
              fontWeight: FontWeight.bold),
        ),
```

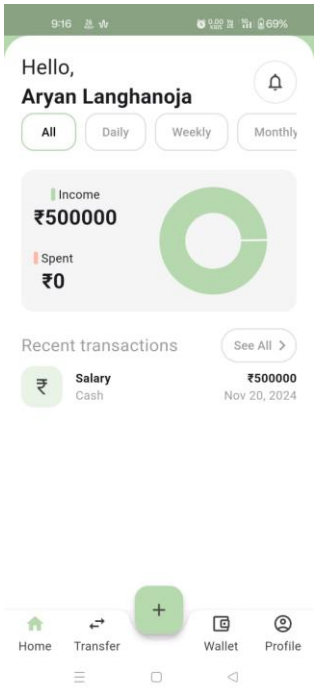
```
                value: label,
                groupValue: controller.paymentType.value,
                activeColor: const Color(0xFFb6d8ae),
                contentPadding: const EdgeInsets.symmetric(horizontal: 10),
                onChanged: (value) {
                  controller.paymentType.value = value!;
                },
              ),
            );
          }
        }
```

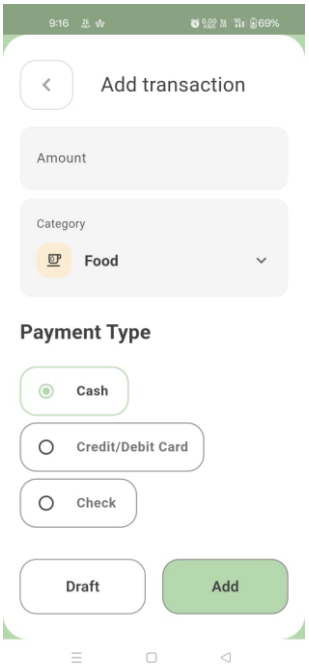
- **main.dart**

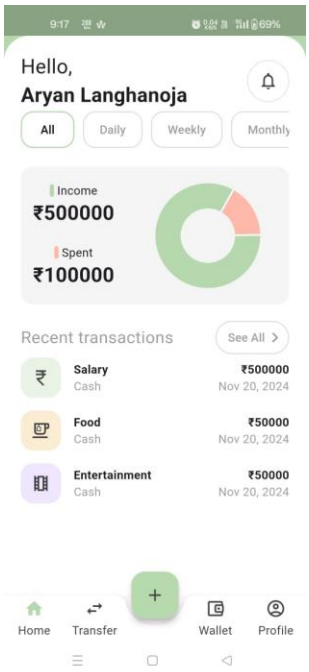
```
import 'package:expense_manager/screen/home_page.dart';
import 'package:flutter/material.dart';
import 'package:get/get.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return GetMaterialApp(
      title: 'Flutter Demo',
      theme: ThemeData(
          // colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
          // useMaterial3: true,
          ),
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
    );
  }
}
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** Daily Expense app using getx.	
Experiment No:- 06	**Date:-** 18-10-2024	**Enrollment No:-** 92200133030

Output:-







	Marwadi University **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** Daily Expense app using getx.	
Experiment No:- 06	**Date:-** 18-10-2024	**Enrollment No:-** 92200133030

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** Shared preference to skip the introduction screens after the first launch.	
Experiment No:- 07	**Date:-** 11-11-2024	**Enrollment No:-** 92200133030

Objective:- Shared preference to skip the introduction screens after the first launch.

Code :-

- **lib**
  - **screen**
    - **dashboard.dart**

```
import 'package:flutter/material.dart';

class DashboardPage extends StatefulWidget {
  const DashboardPage({super.key});

  @override
  State<DashboardPage> createState() => _DashboardPageState();
}

class _DashboardPageState extends State<DashboardPage> {
  List bookList = [
    [
      "The Wager",
      "https://m.media-amazon.com/images/I/81iR3k0Wg6L._SL1500_.jpg"
    ],
    [
      "Atomic habits",
      "https://m.media-amazon.com/images/I/91bYsX41DVL._SL1500_.jpg"
    ],
    ["Dog Man", "https://m.media-amazon.com/images/I/81XqX-wnkIL._SL1500_.jpg"],
    ["Outlive", "https://m.media-amazon.com/images/I/71gpe7LeGSL._SL1500_.jpg"],
  ];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: const Color.fromRGBO(209, 215, 238, 1),
      body: SafeArea(
        child: Container(
          margin: const EdgeInsets.all(10),
          padding: const EdgeInsets.fromLTRB(15, 20, 0, 0),
          decoration: BoxDecoration(
            color: Colors.white,
            borderRadius: BorderRadius.circular(30),
          ),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Row(
```

```
    children: [
      const Expanded(
        child: Text(
          "Hi, Aryan Langhanoja",
          style:
              TextStyle(fontSize: 22, fontWeight: FontWeight.bold),
        ),
      ),
      IconButton(
          onPressed: () {},
          icon: const Icon(
            Icons.search,
            color: Colors.black,
          )),
      IconButton(
        onPressed: () {},
        icon: const Icon(
          Icons.shopping_bag_outlined,
          color: Colors.black,
        ),
      )
    ],
  ),
  const SizedBox(height: 15),
  Expanded(
    child: ListView(
      children: [
        SingleChildScrollView(
          scrollDirection: Axis.horizontal,
          child: Row(
            children: [
              card1(const Color(0xFFd5fffd)),
              card1(const Color(0xFFdce6ff)),
            ],
          ),
        ),
        const SizedBox(height: 30),
        const Row(
          children: [
            Expanded(
                child: Text(
              "Continue Reading",
              style: TextStyle(
                  fontSize: 20, fontWeight: FontWeight.bold),
            )),
            Text(
              "See More",
              style: TextStyle(
                  fontSize: 16, fontWeight: FontWeight.w500),
            ),
            SizedBox(width: 15),
          ],
```

```
                    ),
                    SizedBox(
                      height: 186,
                      child: ListView.builder(
                        scrollDirection: Axis.horizontal,
                        itemCount: bookList.length,
                        itemBuilder: (context, index) =>
                            card2(bookList[index][0], bookList[index][1]),
                      ),
                    ),
                    const SizedBox(height: 10),
                    const Text(
                      "Book of the Day",
                      style:
                          TextStyle(fontSize: 20, fontWeight: FontWeight.bold),
                    ),
                    Container(
                      padding: const EdgeInsets.all(15),
                      margin: const EdgeInsets.fromLTRB(0, 15, 15, 15),
                      decoration: BoxDecoration(
                        color: const Color(0xFFf3f2f3),
                        borderRadius: BorderRadius.circular(15),
                      ),
                      child: Row(
                        children: [
                          Container(
                            decoration: BoxDecoration(
                                border:
                                    Border.all(color: Colors.blue, width: 2),
                                borderRadius: BorderRadius.circular(10)),
                            child: ClipRRect(
                              borderRadius: BorderRadius.circular(10),
                              child: Image.network(
                                "https://m.media-
amazon.com/images/I/81FummIc2eL._SL1500_.jpg",
                                width: 150,
                                height: 150,
                                fit: BoxFit.fitWidth,
                              ),
                            ),
                          ),
                          const SizedBox(width: 10),
                          const Column(
                            crossAxisAlignment: CrossAxisAlignment.start,
                            children: [
                              Text(
                                "It Starts with us",
                                style: TextStyle(
                                    fontSize: 18, fontWeight: FontWeight.bold),
                              ),
                              SizedBox(
                                height: 3,
```

```
                ),
                Text(
                  "Colleen Hoover",
                  style: TextStyle(
                      fontSize: 16, fontWeight: FontWeight.w500),
                ),
                SizedBox(
                  height: 3,
                ),
                Row(
                  children: [
                    Icon(
                      Icons.star,
                      color: Colors.orange,
                      size: 18,
                    ),
                    Icon(
                      Icons.star,
                      color: Colors.orange,
                      size: 18,
                    ),
                    Icon(
                      Icons.star,
                      color: Colors.orange,
                      size: 18,
                    ),
                    Icon(
                      Icons.star,
                      color: Colors.orange,
                      size: 18,
                    ),
                    Icon(
                      Icons.star,
                      color: Colors.orange,
                      size: 18,
                    ),
                  ],
                ),
                SizedBox(
                  height: 10,
                ),
                Text(
                  "\$40  \$60",
                  style: TextStyle(
                      fontSize: 18, fontWeight: FontWeight.bold),
                )
              ],
            ),
          ],
        ),
      ),
    const SizedBox(height: 10),
```

```
                    const Row(
                      children: [
                        Expanded(
                          child: Text(
                            "This Week",
                            style: TextStyle(
                                fontSize: 20, fontWeight: FontWeight.bold),
                          ),
                        ),
                        Text(
                          "See More",
                          style: TextStyle(
                              fontSize: 16, fontWeight: FontWeight.w500),
                        ),
                        SizedBox(width: 15),
                      ],
                    ),
                    Row(
                      children: [
                        card3(
                            "https://m.media-amazon.com/images/I/81Fw7hPAROL._SL1500_.jpg",
                            "Hello Beautiful"),
                        card3(
                            "https://m.media-amazon.com/images/I/818B6S6B-bL._SL1500_.jpg",
                            "Daisy Jones"),
                      ],
                    ),
                  ],
                ),
              ),
            ],
          ),
        ),
      ),
    );
  }

  card1(Color backColor) {
    return Container(
      width: MediaQuery.of(context).size.width * 0.7,
      padding: const EdgeInsets.fromLTRB(15, 20, 15, 20),
      margin: const EdgeInsets.fromLTRB(0, 0, 10, 0),
      decoration: BoxDecoration(
          color: backColor, borderRadius: BorderRadius.circular(10)),
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          const Text(
            "Get Ramadan Offer 2023",
            style: TextStyle(
              fontWeight: FontWeight.w500,
              fontSize: 16,
```

```
          ),
        ),
        const SizedBox(height: 5),
        const Text(
          "25% on All Islamic Books",
          style: TextStyle(
            fontWeight: FontWeight.bold,
            fontSize: 18,
          ),
        ),
        const SizedBox(height: 15),
        ElevatedButton(
          onPressed: () {},
          style: ElevatedButton.styleFrom(
            shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(10),
            ),
            fixedSize: const Size.fromWidth(130),
            backgroundColor: Colors.black,
            padding: const EdgeInsets.all(16),
          ),
          child: const Text(
            "Read More",
            style: TextStyle(
              color: Colors.white, // Make sure the text color is contrasting
            ),
          ),
        ),
      ],
    ),
  );
}

card2(String bookName, String imgPath) {
  return Container(
    margin: const EdgeInsets.fromLTRB(0, 15, 10, 10),
    width: 90,
    child: Column(
      children: [
        Stack(
          alignment: Alignment.topLeft,
          children: [
            ClipRRect(
              borderRadius: BorderRadius.circular(10),
              child: Image.network(
                imgPath,
                width: 90,
              ),
            ),
            Container(
              decoration: BoxDecoration(
                color: Colors.white,
```

```
                  borderRadius: BorderRadius.circular(30),
                ),
                width: 71,
                padding: const EdgeInsets.fromLTRB(10, 3, 10, 3),
                margin: EdgeInsets.fromLTRB(5, 5, 0, 0),
                child: const Row(
                  children: [
                    Icon(
                      Icons.bolt,
                      size: 16,
                    ),
                    Text(
                      "Stream",
                      style:
                          TextStyle(fontSize: 10, fontWeight: FontWeight.bold),
                    )
                  ],
                ),
              )
            ],
          ),
          Text(bookName, overflow: TextOverflow.ellipsis),
        ],
      ),
    );
  }

  card3(String imgPath, String bookName) {
    return Container(
      margin: const EdgeInsets.fromLTRB(0, 10, 0, 10),
      width: (MediaQuery.of(context).size.width - 35) * 0.5,
      child: Row(
        children: [
          ClipRRect(
            borderRadius: BorderRadius.circular(10),
            child: Image.network(
              imgPath,
              width: (MediaQuery.of(context).size.width - 35) * 0.22,
            ),
          ),
          const SizedBox(width: 8),
          Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Container(
                decoration: BoxDecoration(
                  color: const Color(0xFFfed6c6),
                  borderRadius: BorderRadius.circular(10),
                ),
                padding:
                    const EdgeInsets.symmetric(horizontal: 10, vertical: 2),
                child: const Text(
```

```
                "New Arrival",
                overflow: TextOverflow.ellipsis,
                style: TextStyle(fontSize: 10),
              ),
            ),
            const SizedBox(height: 10),
            SizedBox(
              width: (MediaQuery.of(context).size.width - 35) * 0.24,
              child: Text(
                bookName,
                overflow: TextOverflow.ellipsis,
              ),
            ),
          ],
        )
      ],
    ),
  );
 }
}
```

- **introduction.dart**

```
import 'package:e_book/screens/dashboard.dart';
import 'package:flutter/material.dart';
import 'package:introduction_screen/introduction_screen.dart';
import 'package:shared_preferences/shared_preferences.dart';

class IntroductionPage extends StatefulWidget {
  const IntroductionPage({super.key});

  @override
  State<IntroductionPage> createState() => _IntroductionPageState();
}

class _IntroductionPageState extends State<IntroductionPage> {
  @override
  Widget build(BuildContext context) {
    List<PageViewModel> listPagesViewModel = [
      introPageModel("Welcome", "Welcome to this awesome intro screen app.",
          "assets/images/intro_1.png", const Color.fromRGBO(126, 196, 128, 1)),
      introPageModel(
          "Awesome App",
          "This is an awesome app, of intro screen design",
          "assets/images/intro_2.png",
          const Color.fromRGBO(98, 181, 243, 1)),
      introPageModel("Flutter App", "Flutter is awesome fro app development",
          "assets/images/intro_3.png", const Color.fromRGBO(123, 133, 205, 1)),
    ];
```

```
  return Scaffold(
    body: IntroductionScreen(
      pages: listPagesViewModel,
      showSkipButton: true,
      skip: const Text(
        "Skip",
        style: TextStyle(color: Colors.black),
      ),
      next: const Icon(
        Icons.arrow_forward,
        color: Colors.black,
      ),
      done: const Icon(
        Icons.check,
        color: Colors.black,
      ),
      onDone: () async {
        final SharedPreferences prefs = await SharedPreferences.getInstance();
        await prefs.setBool('firstTime', false);
        // ignore: use_build_context_synchronously
        Navigator.pushReplacement(context,
            MaterialPageRoute(builder: (context) => const DashboardPage()));
      },
      onSkip: () async {
        final SharedPreferences prefs = await SharedPreferences.getInstance();
        await prefs.setBool('firstTime', false);
        // ignore: use_build_context_synchronously
        Navigator.pushReplacement(context,
            MaterialPageRoute(builder: (context) => const DashboardPage()));
      },
      controlsPadding: const EdgeInsets.fromLTRB(0, 16, 0, 16),
      dotsDecorator: DotsDecorator(
        size: const Size.square(10),
        activeSize: const Size(20, 20),
        activeColor: Colors.red,
        color: Colors.grey[300]!,
        spacing: const EdgeInsets.symmetric(horizontal: 3.0),
        activeShape:
            RoundedRectangleBorder(borderRadius: BorderRadius.circular(25.0)),
      ),
    ),
  );
}

introPageModel(
    String title, String bodyText, String imgPath, Color backColor) {
  return PageViewModel(
    title: title,
    body: bodyText,
    image: Container(
      width: MediaQuery.of(context).size.width * 0.9,
      height: 500,
```

```
        padding: const EdgeInsets.fromLTRB(0, 25, 0, 25),
        decoration: BoxDecoration(
          color: Colors.white,
          borderRadius: BorderRadius.circular(25),
        ),
        child: Image.asset(imgPath, fit: BoxFit.contain),
      ),
      reverse: true,
      decoration: PageDecoration(
        pageMargin: const EdgeInsets.all(0),
        titlePadding: const EdgeInsets.fromLTRB(0, 100, 0, 0),
        bodyPadding: const EdgeInsets.fromLTRB(0, 30, 0, 0),
        imagePadding: const EdgeInsets.fromLTRB(0, 0, 0, 100),
        pageColor: backColor,
        titleTextStyle: const TextStyle(
            fontSize: 42, color: Colors.white, fontWeight: FontWeight.bold),
        bodyTextStyle: const TextStyle(fontSize: 28, color: Colors.white),
      ),
    );
  }
}
```

- ## splash_screen.dart

```
import 'dart:async';

import 'package:e_book/screens/dashboard.dart';
import 'package:e_book/screens/introduction.dart';
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

class SplashScreen extends StatefulWidget {
  const SplashScreen({super.key});

  @override
  State<SplashScreen> createState() => _SplashScreenState();
}

class _SplashScreenState extends State<SplashScreen> {
  @override
  void initState() {
    // TODO: implement initState
    super.initState();
    Timer(const Duration(seconds: 3), () async {
      // Obtain shared preferences.
      final SharedPreferences prefs = await SharedPreferences.getInstance();
      final bool firstTime = prefs.getBool('firstTime') ?? true;
      if (firstTime) {
        // ignore: use_build_context_synchronously
```

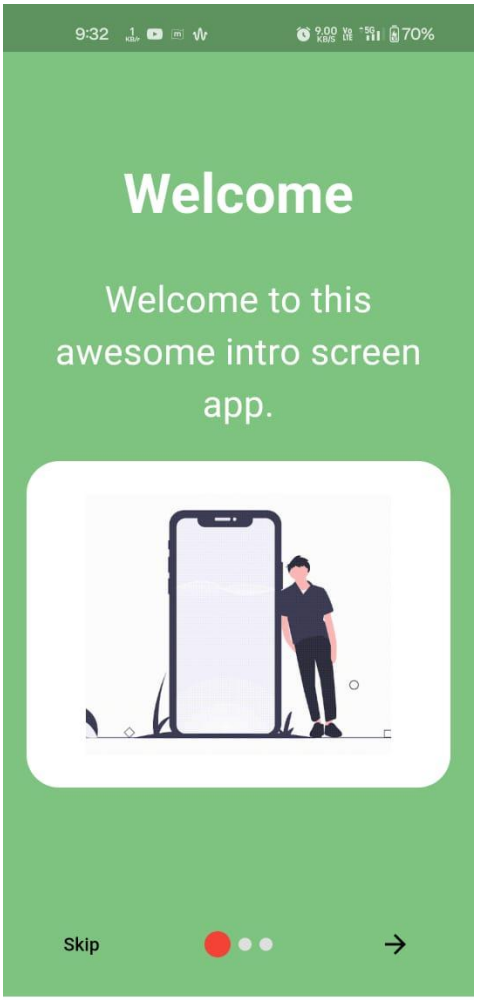
```
          Navigator.pushReplacement(context,
              MaterialPageRoute(builder: (context) => const IntroductionPage()));
        }
        else {
          // ignore: use_build_context_synchronously
          Navigator.pushReplacement(context,
              MaterialPageRoute(builder: (context) => const DashboardPage()));
        }
      });
    }

    @override
    Widget build(BuildContext context) {
      return Scaffold(
        body: Center(
          child: Container(
            height: 200,
            width: 200,
            alignment: Alignment.center,
            decoration: BoxDecoration(
              color: Colors.grey[300],
              borderRadius: BorderRadius.circular(30),
            ),
            child: const Text(
              "E-Book",
              style: TextStyle(fontSize: 32, fontWeight: FontWeight.bold),
            ),
          ),
        ),
      );
    }
  }
```

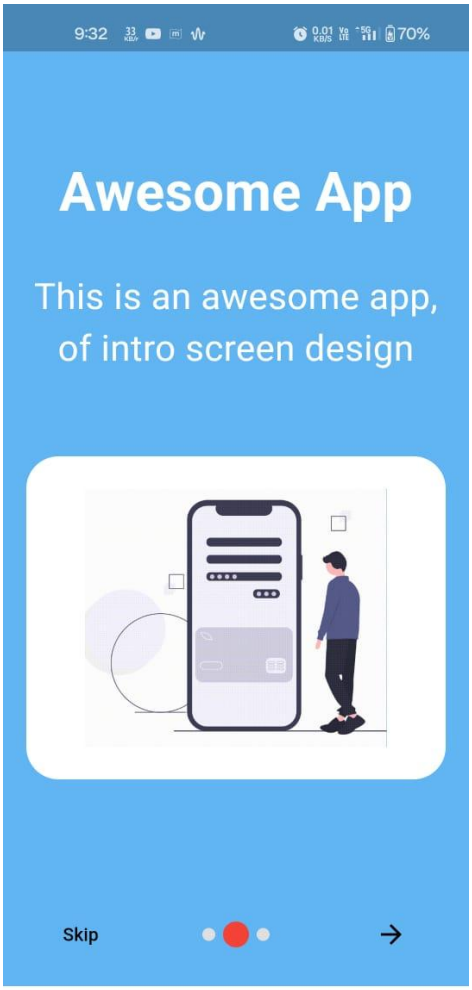
- **main.dart**

```
import 'package:e_book/screens/splash_screen.dart';
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  // This widget is the root of your application.
  @override
```

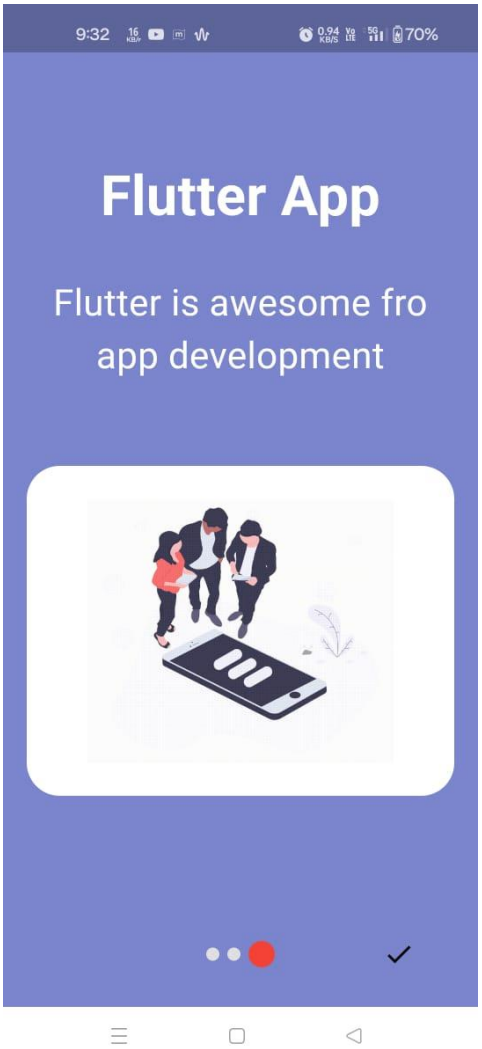
```
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Demo',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
        useMaterial3: true,
      ),
      debugShowCheckedModeBanner: false,
      home: const SplashScreen(),
    );
  }
}
```

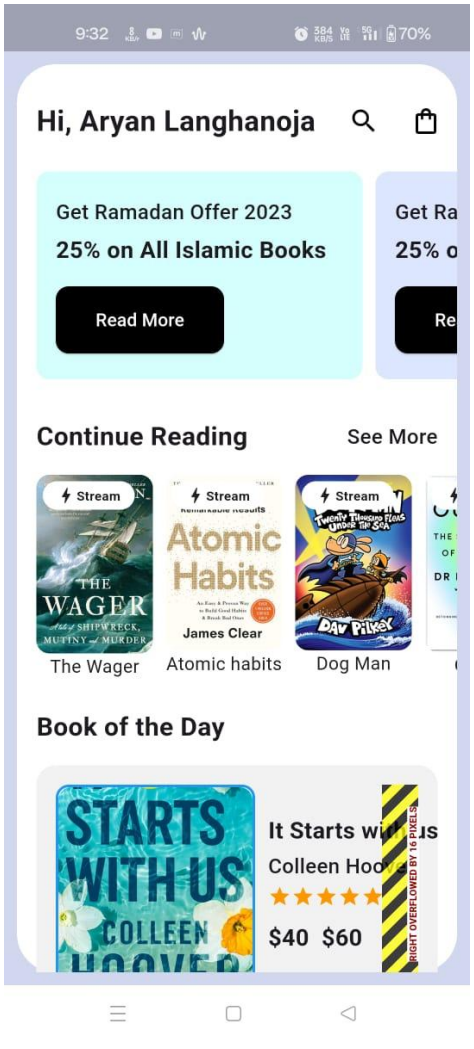
Output:-





Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** Shared preference to skip the introduction screens after the first launch.
Experiment No:- 07	**Date:-** 11-11-2024 / **Enrollment No:-** 92200133030





Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** Blog Application using Firebase	
Experiment No:- 10	**Date:-** 20-11-2024	**Enrollment No:-** 92200133030

Objective:- Blog Application using Firebase

Code :-

- **Lib**
  - **models**
    - **blog.dart**

```
class Blog {
  String? title;
  String? description;
  String? id;
  Blog({this.title, this.description, this.id});
}
```

  - **services**
    - **firebase_services.dart**

```
import 'package:cloud_firestore/cloud_firestore.dart';
import '../models/blog.dart';

class FirebaseServices {
  var db = FirebaseFirestore.instance;
  addBlog(String title, String description) {
    // Create a new note with a title and description
    final note = <String, dynamic>{"title": title, "description": description};

// Add a new document with a generated ID
    db.collection("blog").add(note).then((DocumentReference doc) =>
        // ignore: avoid_print
        print('DocumentSnapshot added with ID: ${doc.id}'));
  }

  Stream<QuerySnapshot> fetchBlog() {
    Stream<QuerySnapshot> collectionStream =
        FirebaseFirestore.instance.collection('blog').snapshots();
    return collectionStream;
// Stream documentStream =
FirebaseFirestore.instance.collection('users').doc('ABC123').snapshots();
  }

  updateBlog(Blog blog) {
    return db
        .collection("notes")
```

```
        .doc(blog.id)
        .update({'title': blog.title, 'description': blog.description})
        // ignore: avoid_print
        .then((value) => print("Note Updated"))
        // ignore: avoid_print
        .catchError((error) => print("Failed to update Note: $error"));
  }

  deleteBlog(id) {
    return db
        .collection("blog")
        .doc(id)
        .delete()
        // ignore: avoid_print
        .then((value) => print("User Deleted"))
        // ignore: avoid_print
        .catchError((error) => print("Failed to delete user: $error"));
  }
}
```

- **screen**
  - **blog_details_screen.dart**

```
import 'package:flutter/material.dart';

import '../models/blog.dart';
import '../services/firebase_services.dart';

class BlogDetailsScreen extends StatefulWidget {
  final Blog? blog;
  const BlogDetailsScreen({super.key, this.blog});
  @override
  State<BlogDetailsScreen> createState() => _BlogDetailsScreenState();
}

class _BlogDetailsScreenState extends State<BlogDetailsScreen> {
  final _formKey = GlobalKey<FormState>();
  late String _title;
  late String _content;

  @override
  void initState() {
    super.initState();
    if (widget.blog != null) {
      _title = widget.blog!.title!;
      _content = widget.blog!.description!;
    } else {
      _title = '';
      _content = '';
    }
```

```
  }

  void _saveBlog() {
    if (_formKey.currentState!.validate()) {
      _formKey.currentState!.save();
      // final blog = Blog(
      //   id: widget.blog?.id,
      //   title: _title,
      //   description: _content,
      // );
//todo code to store on firestore
      FirebaseServices().addBlog(_title, _content);
      Navigator.pop(context);
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text(widget.blog == null ? 'New Blog' : 'Edit Blog'),
      ),
      body: Padding(
        padding: const EdgeInsets.all(16.0),
        child: Form(
          key: _formKey,
          child: Column(
            children: [
              TextFormField(
                initialValue: _title,
                decoration: const InputDecoration(labelText: 'Title'),
                validator: (value) {
                  if (value == null || value.isEmpty) {
                    return 'Please enter a title';
                  }
                  return null;
                },
                onSaved: (value) {
                  _title = value!;
                },
              ),
              const SizedBox(height: 16.0),
              TextFormField(
                initialValue: _content,
                decoration: const InputDecoration(labelText: 'Content'),
                validator: (value) {
                  if (value == null || value.isEmpty) {
                    return 'Please enter content';
                  }
                  return null;
                },
                onSaved: (value) {
```

```
                _content = value!;
              },
            ),
            const SizedBox(height: 16.0),
            ElevatedButton(
              onPressed: _saveBlog,
              child: const Text('Save'),
            ),
          ],
        ),
      ),
    ),
  );
 }
}
```

- **blog_list_screen.dart**

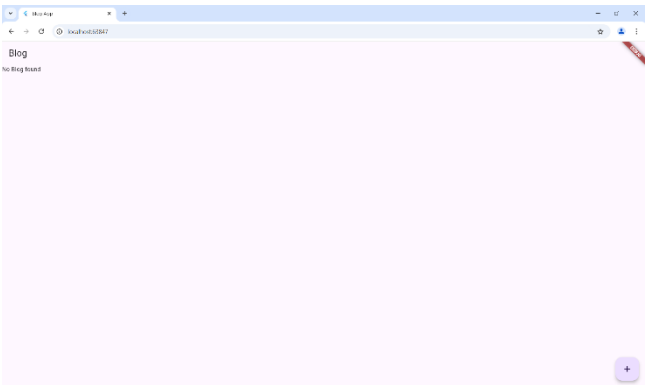
```
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:flutter/material.dart';
import 'package:blog_app/services/firebase_services.dart';

import '../models/blog.dart';
import 'blog_details_screen.dart';

class BlogListScreen extends StatelessWidget {
  const BlogListScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Blog'),
      ),
      body: StreamBuilder<QuerySnapshot>(
        stream: FirebaseServices().fetchBlog(),
        builder: (BuildContext context, AsyncSnapshot<QuerySnapshot> snapshot) {
          if (snapshot.hasError) {
            return const Text('Something went wrong');
          }

          if (snapshot.connectionState == ConnectionState.waiting) {
            return const Center(child: CircularProgressIndicator());
          }

          if (!snapshot.hasData ||
              snapshot.data == null ||
              snapshot.data!.docs.isEmpty) {
            return const Text("No Blog found");
```

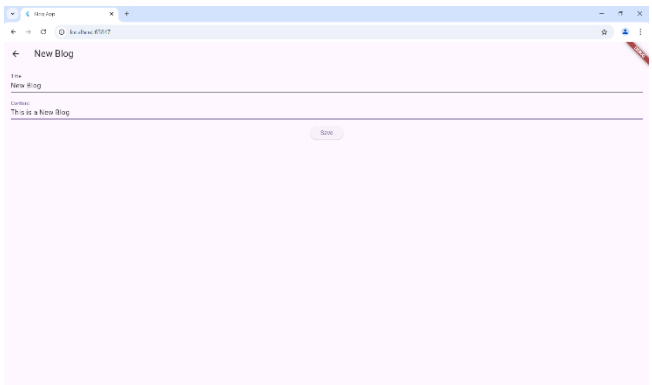
```
}

return ListView(
  children: snapshot.data!.docs.map((DocumentSnapshot document) {
    Map<String, dynamic>? data =
        document.data() as Map<String, dynamic>?;

    if (data == null) {
      return const ListTile(
        title: Text('No Data'),
      );
    }

    // Safely handle null for title and description
    String title = data['title'] ?? 'Untitled';
    String description = data['description'] ?? 'No description';

    Blog blog = Blog(
      id: document.id,
      title: title,
      description: description,
    );

    return ListTile(
      title: Text(blog.title!),
      subtitle: Text(blog.description!),
      trailing: Row(
        mainAxisSize: MainAxisSize.min,
        children: [
          IconButton(
            icon: const Icon(Icons.edit, color: Colors.blue),
            onPressed: () {
              // Navigate to edit blog screen
              Navigator.of(context).push(MaterialPageRoute(
                builder: (context) => BlogDetailsScreen(blog: blog),
              ));
            },
          ),
          IconButton(
            icon: const Icon(Icons.delete, color: Colors.red),
            onPressed: () {
              // Confirm deletion before deleting
              showDialog(
                context: context,
                builder: (BuildContext context) {
                  return AlertDialog(
                    title: const Text("Delete Blog"),
                    content: const Text(
                        "Are you sure you want to delete this blog?"),
                    actions: [
                      TextButton(
                        onPressed: () => Navigator.of(context).pop(),
```

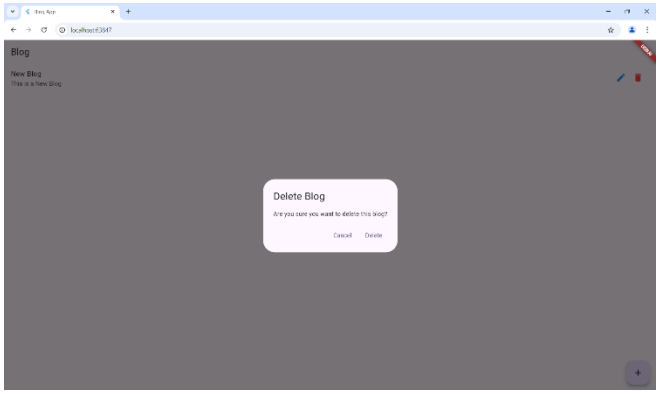
```
                                    child: const Text("Cancel"),
                                  ),
                                  TextButton(
                                    onPressed: () {
                                      FirebaseServices().deleteBlog(document.id);
                                      Navigator.of(context).pop();
                                    },
                                    child: const Text("Delete"),
                                  ),
                                ],
                              );
                            },
                          );
                        },
                      ),
                    ],
                  ),
                );
              }).toList(),
            );
          },
        ),
        floatingActionButton: FloatingActionButton(
          onPressed: () {
            Navigator.of(context).push(MaterialPageRoute(
                builder: (context) => const BlogDetailsScreen()));
          },
          child: const Icon(Icons.add),
        ),
      );
    }
  }
```

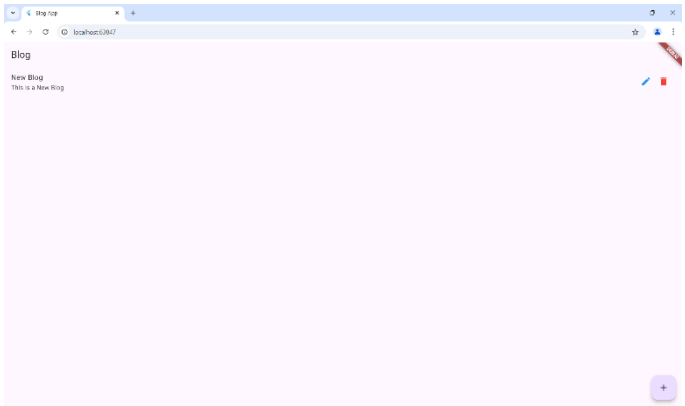
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** Blog Application using Firebase	
Experiment No:- 10	**Date:-** 20-11-2024	**Enrollment No:-** 92200133030

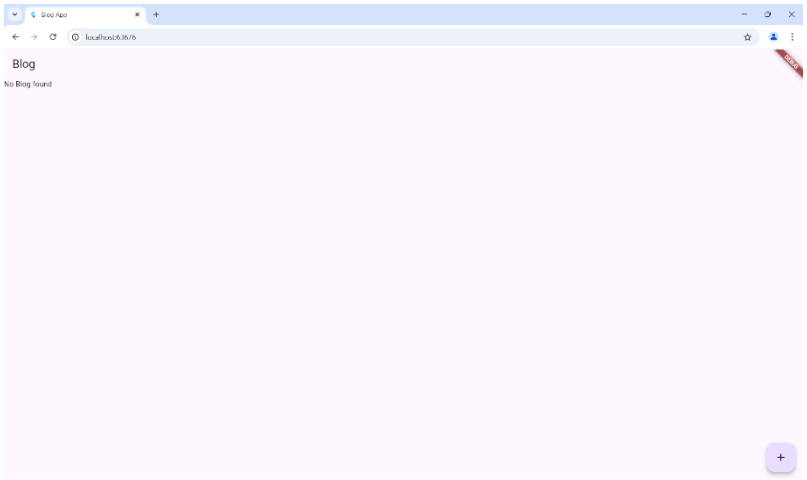
Output:-











Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Cross-Platform Mobile Application Development (01CT1517)	**Aim:** Create Notes app and build for web deploy it on GitHub pages also create release APK and appbundle.	
Experiment No:- 10	**Date:-** 20-11-2024	**Enrollment No:-** 92200133030

Objective:- Create Notes app and build for web deploy it on GitHub pages also create release APK and appbundle.

Code :-

- **Lib**

  - **models**
    - **notes.dart**

```
class Note {
  String? title;
  String? description;
  String? id;
  Note({this.title, this.description, this.id});
}
```

  - **services**
    - **firebase_services.dart**

```
import 'package:cloud_firestore/cloud_firestore.dart';
import '../models/notes.dart';

class FirebaseServices {
  var db = FirebaseFirestore.instance;
  addNote(String title, String description) {
    // Create a new note with a title and description
    final note = <String, dynamic>{"title": title, "description": description};

// Add a new document with a generated ID
    db.collection("notes").add(note).then((DocumentReference doc) =>
        print('DocumentSnapshot added with ID: ${doc.id}'));
  }

  Stream<QuerySnapshot> fetchNotes() {
    Stream<QuerySnapshot> collectionStream =
        FirebaseFirestore.instance.collection('notes').snapshots();
    return collectionStream;
// Stream documentStream =
FirebaseFirestore.instance.collection('users').doc('ABC123').snapshots();
  }

  updateNote(Note note) {
    return db
        .collection("notes")
```

```
                .doc(note.id)
                .update({'title': note.title, 'description': note.description})
                .then((value) => print("Note Updated"))
                .catchError((error) => print("Failed to update Note: $error"));
        }

        deleteNote(id) {
          return db
                .collection("notes")
                .doc(id)
                .delete()
                .then((value) => print("User Deleted"))
                .catchError((error) => print("Failed to delete user: $error"));
        }
      }
```

- **screen**
  - **notes_details_screen.dart**

```
import 'package:flutter/material.dart';

import '../models/notes.dart';
import '../services/firebase_services.dart';

class NotesDetailScreen extends StatefulWidget {
  final Note? note;
  NotesDetailScreen({this.note});
  @override
  State<NotesDetailScreen> createState() => _NotesDetailScreenState();
}

class _NotesDetailScreenState extends State<NotesDetailScreen> {
  final _formKey = GlobalKey<FormState>();
  late String _title;
  late String _content;

  @override
  void initState() {
    super.initState();
    if (widget.note != null) {
      _title = widget.note!.title!;
      _content = widget.note!.description!;
    } else {
      _title = '';
      _content = '';
    }
  }

  void _saveNote() {
```

```
      if (_formKey.currentState!.validate()) {
        _formKey.currentState!.save();
        final note = Note(
          id: widget.note?.id,
          title: _title,
          description: _content,
        );
//todo code to store on firestore
        FirebaseServices().addNote(_title, _content);
        Navigator.pop(context);
      }
    }

    @override
    Widget build(BuildContext context) {
      return Scaffold(
        appBar: AppBar(
          title: Text(widget.note == null ? 'New Note' : 'Edit Note'),
        ),
        body: Padding(
          padding: const EdgeInsets.all(16.0),
          child: Form(
            key: _formKey,
            child: Column(
              children: [
                TextFormField(
                  initialValue: _title,
                  decoration: InputDecoration(labelText: 'Title'),
                  validator: (value) {
                    if (value == null || value.isEmpty) {
                      return 'Please enter a title';
                    }
                    return null;
                  },
                  onSaved: (value) {
                    _title = value!;
                  },
                ),
                SizedBox(height: 16.0),
                TextFormField(
                  initialValue: _content,
                  decoration: InputDecoration(labelText: 'Content'),
                  validator: (value) {
                    if (value == null || value.isEmpty) {
                      return 'Please enter content';
                    }
                    return null;
                  },
                  onSaved: (value) {
                    _content = value!;
                  },
                ),
```

```
                    SizedBox(height: 16.0),
                    ElevatedButton(
                      onPressed: _saveNote,
                      child: Text('Save'),
                    ),
                  ],
                ),
              ),
            ),
          ),
        );
      }
    }
```

- **notes _list_screen.dart**

```
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:flutter/material.dart';
import 'package:notes_app/services/firebase_services.dart';

import '../models/notes.dart';
import 'notes_details_screen.dart';

class NotesListScreen extends StatelessWidget {
  const NotesListScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Notes'),
      ),
      body: StreamBuilder<QuerySnapshot>(
        stream: FirebaseServices().fetchNotes(),
        builder: (BuildContext context, AsyncSnapshot<QuerySnapshot> snapshot) {
          if (snapshot.hasError) {
            return Text('Something went wrong');
          }

          if (snapshot.connectionState == ConnectionState.waiting) {
            return Center(child: CircularProgressIndicator());
          }

          if (!snapshot.hasData ||
              snapshot.data == null ||
              snapshot.data!.docs.isEmpty) {
            return Text("No notes found");
          }
```

```
          return ListView(
            children: snapshot.data!.docs.map((DocumentSnapshot document) {
              Map<String, dynamic>? data =
                  document.data() as Map<String, dynamic>?;

              if (data == null) {
                return ListTile(
                  title: Text('No Data'),
                );
              }

              // Safely handle null for title and description
              String title = data['title'] ?? 'Untitled';
              String description = data['description'] ?? 'No description';

              Note note = Note(
                id: document.id,
                title: title,
                description: description,
              );

              return GestureDetector(
                onDoubleTap: () {
                  FirebaseServices().deleteNote(document.id);
                },
                onTap: () {
                  Navigator.of(context).push(MaterialPageRoute(
                      builder: (context) => NotesDetailScreen(note: note)));
                },
                child: ListTile(
                  title: Text(note.title!),
                  subtitle: Text(note.description!),
                ),
              );
            }).toList(),
          );
        },
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: () {
          Navigator.of(context).push(
              MaterialPageRoute(builder: (context) => NotesDetailScreen()));
        },
        child: Icon(Icons.add),
      ),
    );
  }
}
```

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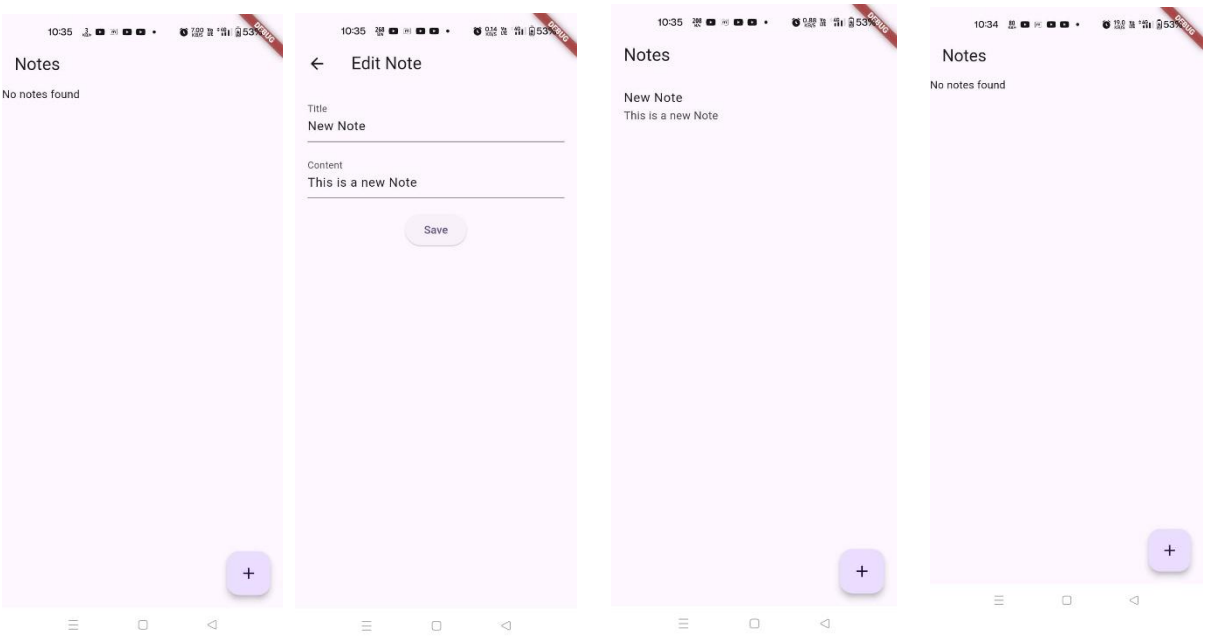
Output:-